

# **CHM 3120: Introduction to Analytical Chemistry**

## **Unit 4 - Section 2:**

# **Atomic Spectroscopy**

## **Chapter 26**

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# Atomic Spectroscopy

In optical atomic spectroscopy techniques, samples are **atomized** by **intense heat** into free atoms/ions of the element(s) of interest.

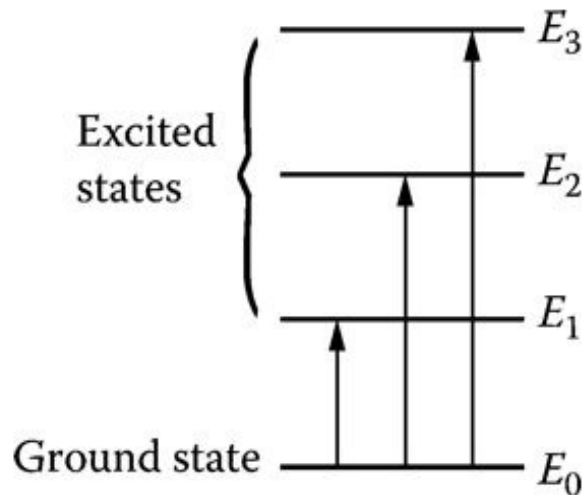
*Different techniques exist, depending on the energy transition occurring:*

- Atomic Absorption Spectroscopy (AAS)
- Atomic Fluorescence Spectroscopy (AFS)
- Atomic Emission Spectroscopy (AES/OES)

# Elemental Transition Lines

In optical techniques, gaseous atoms and ions undergo electronic transitions in which the valence electrons are excited to higher energy levels.

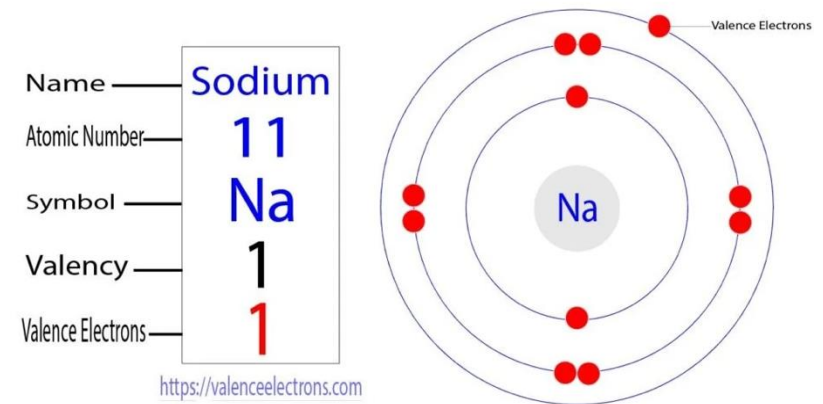
The resulting absorption/emission lines are determined by the elements unique electronic configuration.



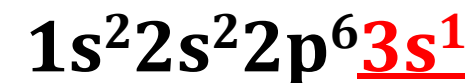
$$\Delta E' = \frac{hc}{\lambda_1} = E_1 - E_0$$

$$\Delta E'' = \frac{hc}{\lambda_2} = E_2 - E_0$$

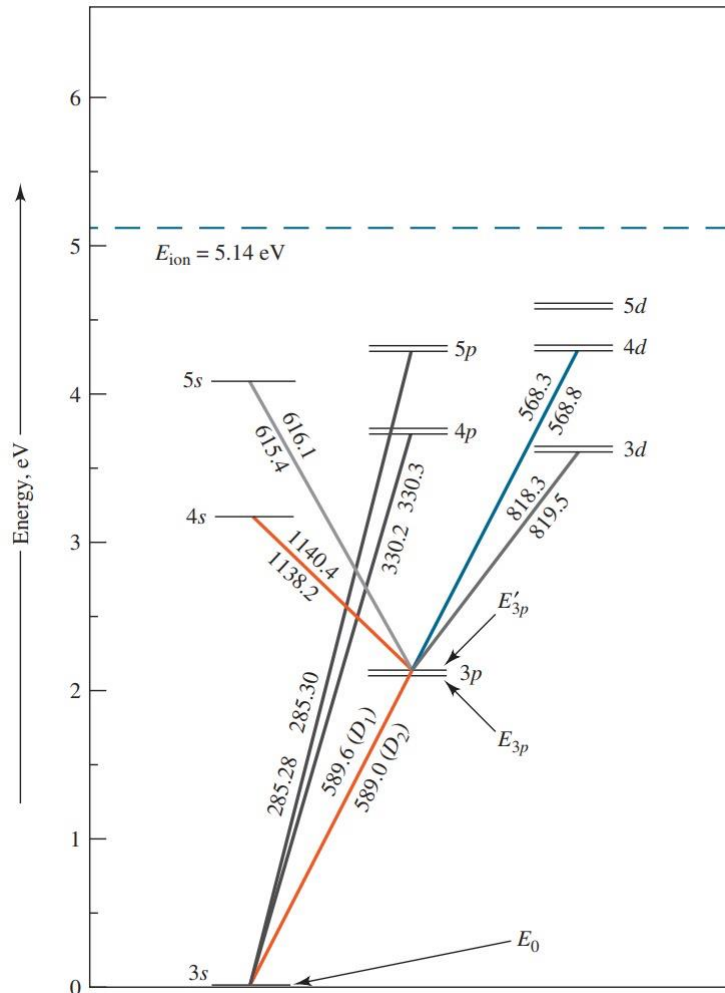
$$\Delta E''' = \frac{hc}{\lambda_3} = E_3 - E_0$$



Ex. Sodium:



# Elemental Energy Level Diagrams

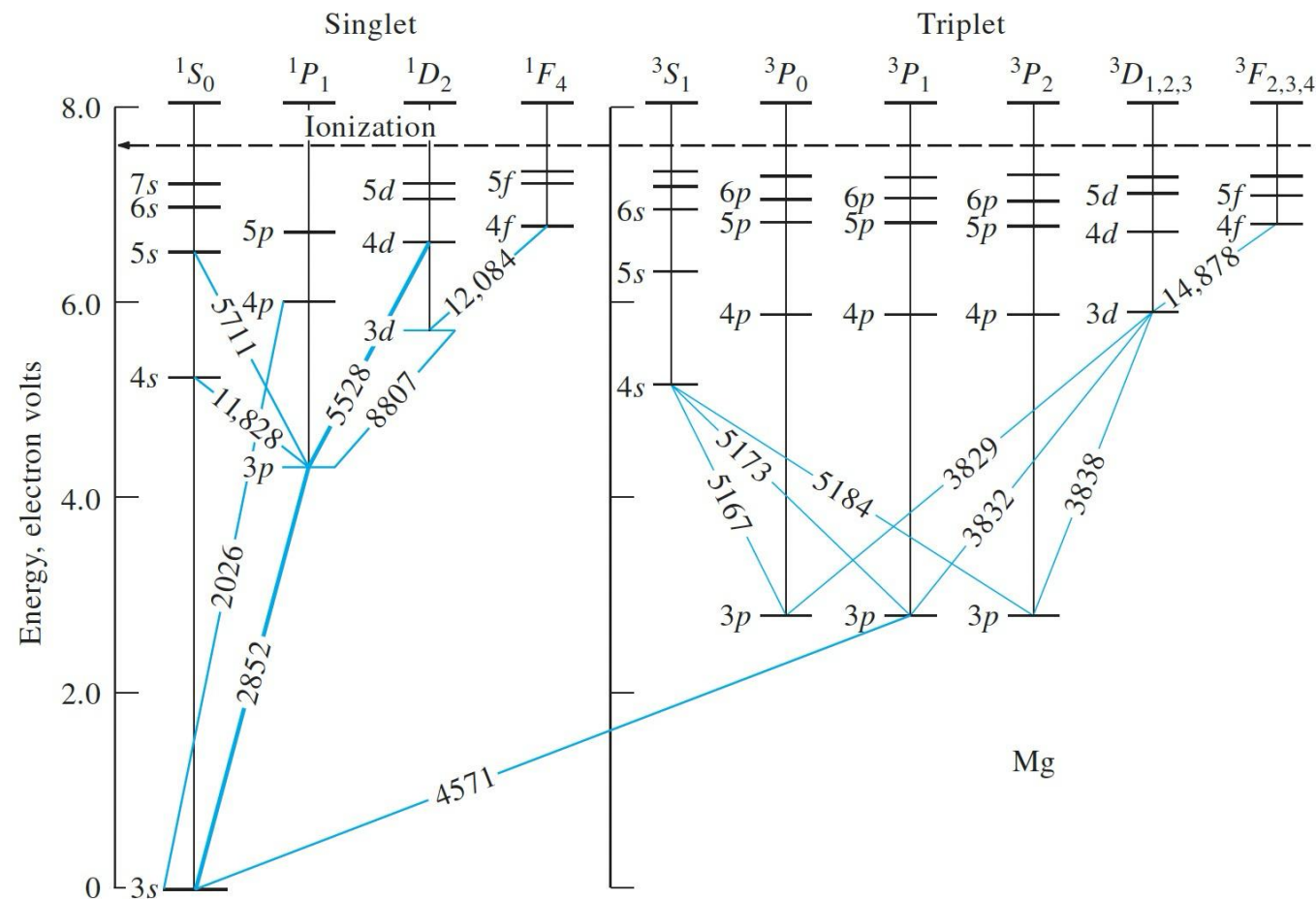
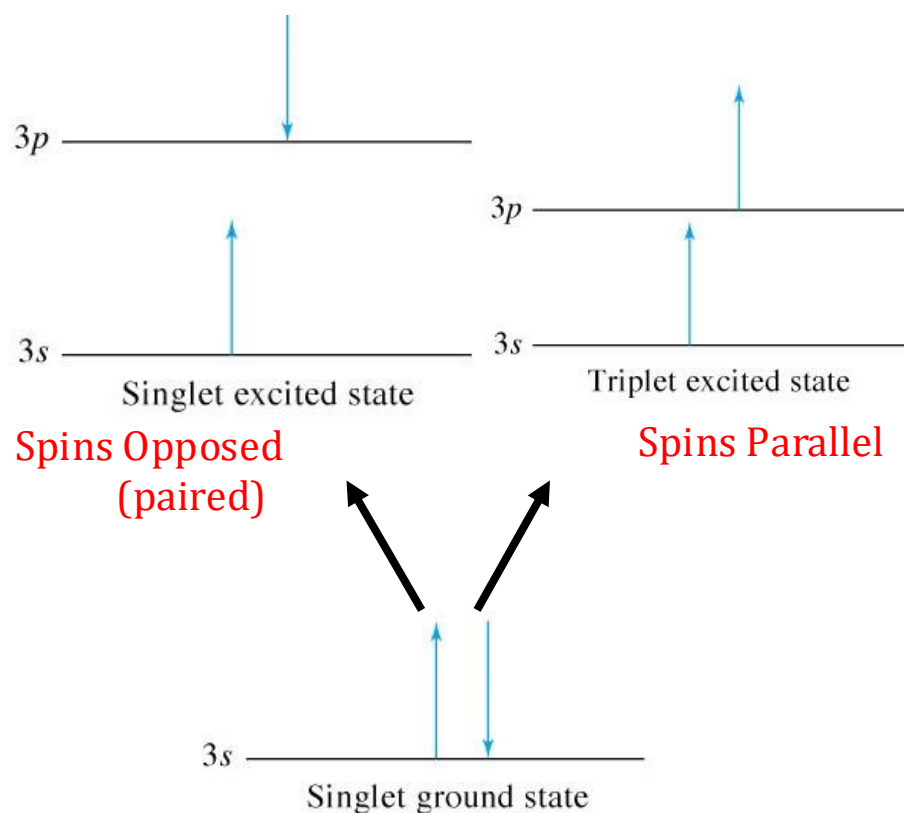


**Grotrian diagrams** are used to show the absorption energy levels of atoms.

- The **3s orbital**, containing the valence electron, is assigned an energy value of zero.
- The scale extends to about 5.14 eV, the energy required to remove the single 3s electron to produce a sodium ion, the **ionization energy**.
- **Orbital splitting** is observed in the higher energy subshells (p, d, f) due to electron spin and orbital motion.
- Both the spin and orbital motion create **magnetic fields** due to the rotation of the electron charge.

# Elemental Energy Level Diagrams

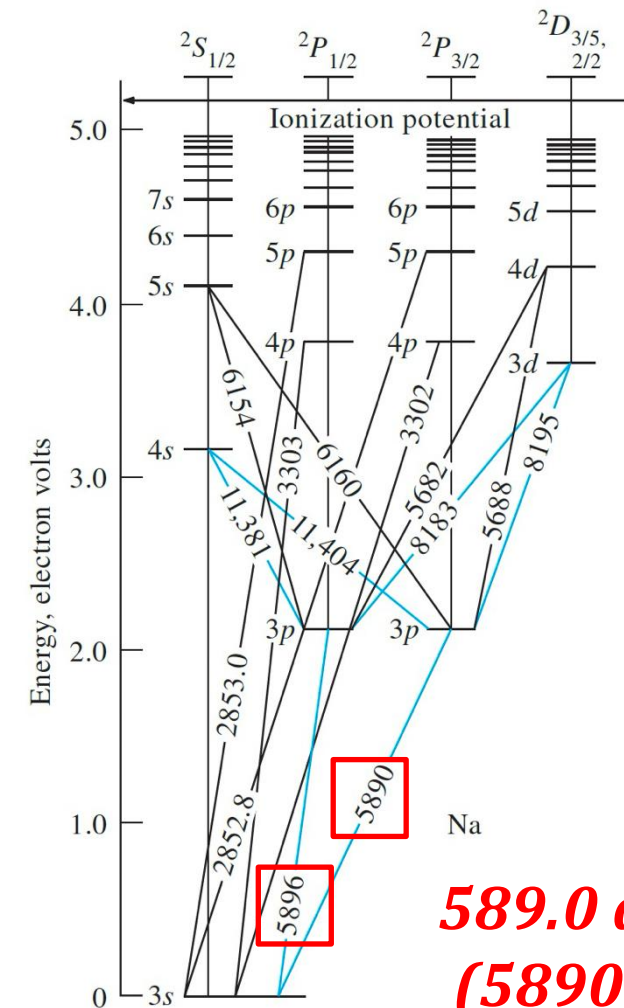
Recall, for elements with two electrons in the s-orbital, an excited **singlet state** and **triplet state** exist.



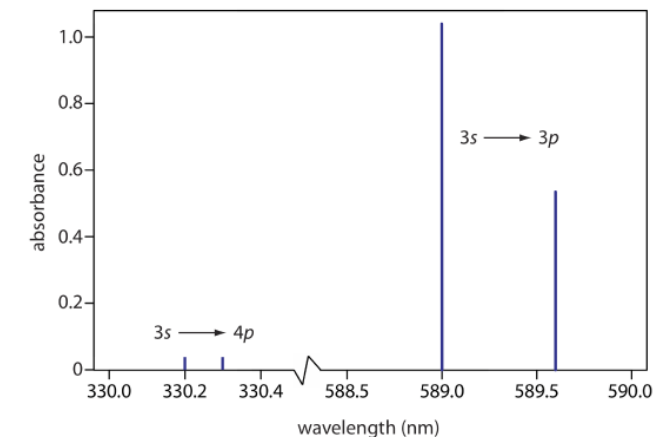
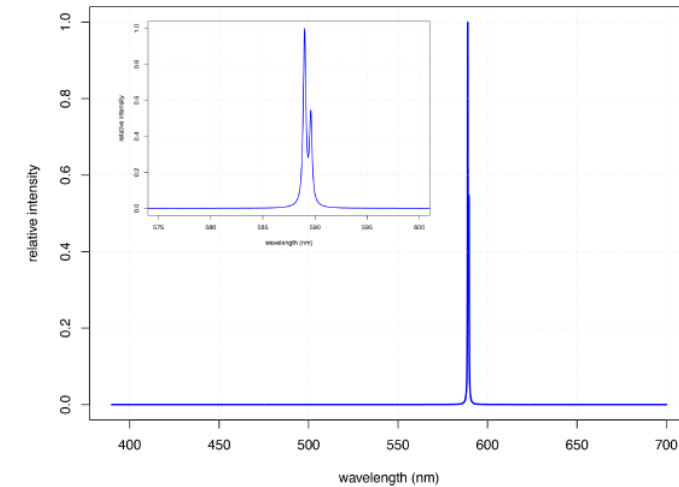
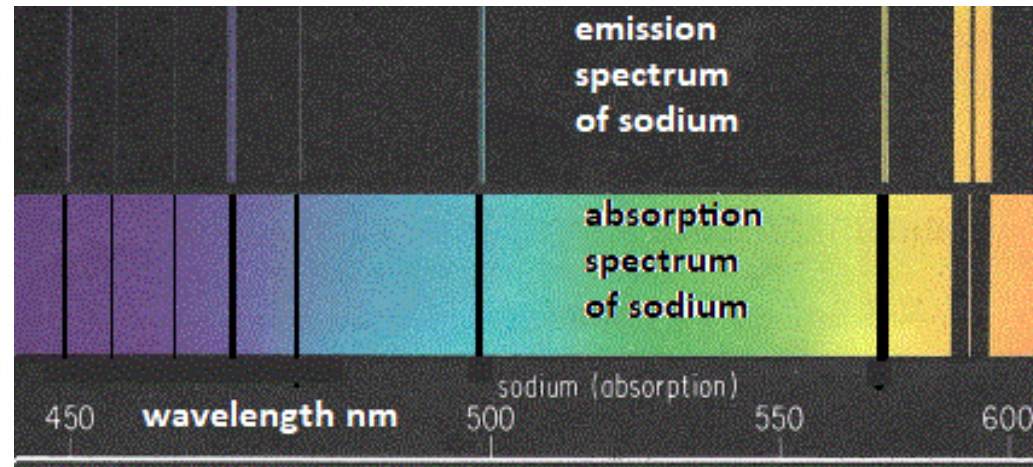
# Absorption and Emission Spectra

The absorption and emission spectra of atoms are generally complimentary.

Transitions that occur between an excited electronic state and the ground state are known as **resonance lines**.

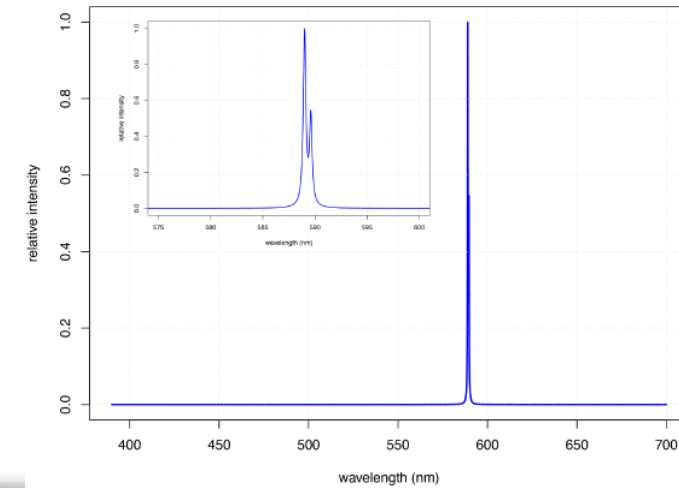
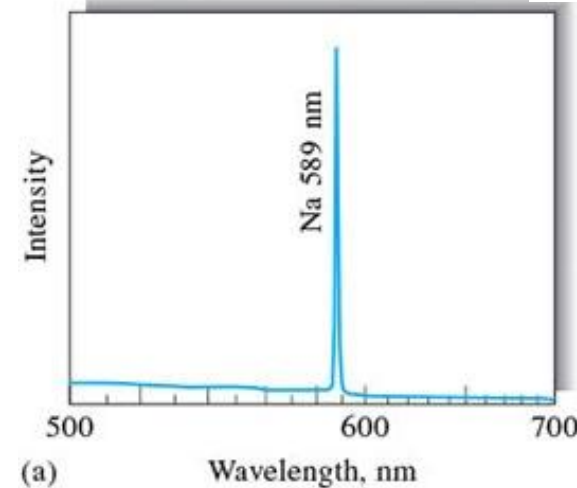
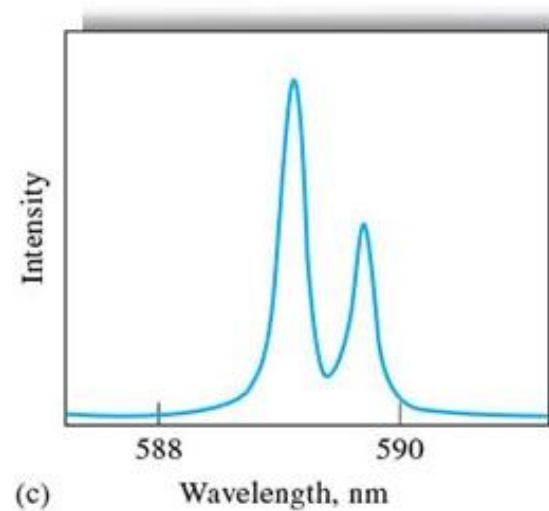
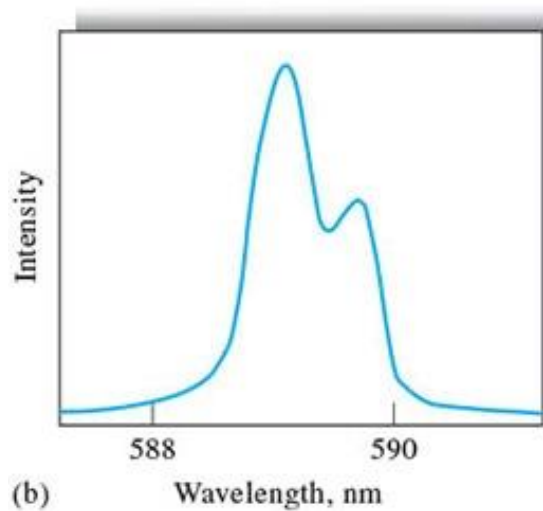


**589.0 and 589.6 nm  
(5890 and 5896 Å)**



# Absorption and Emission Spectra

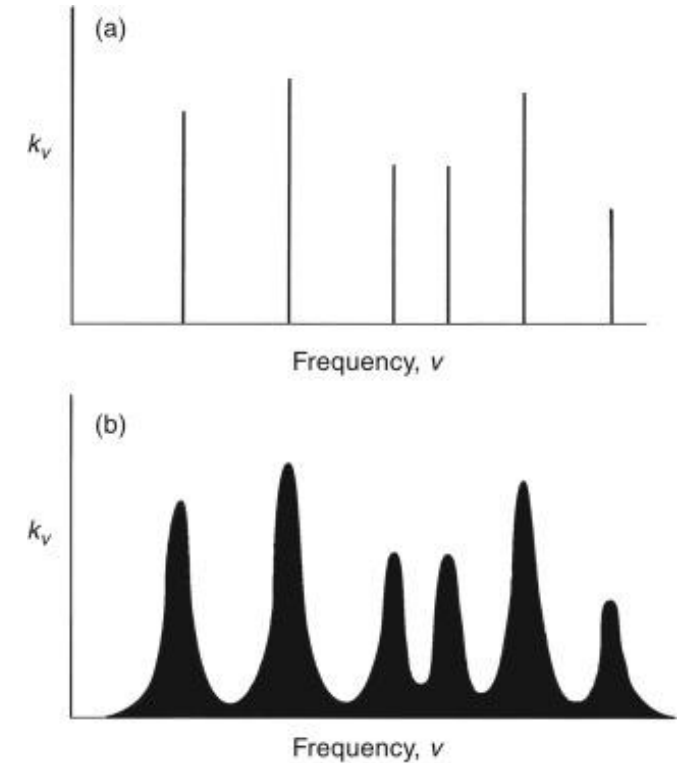
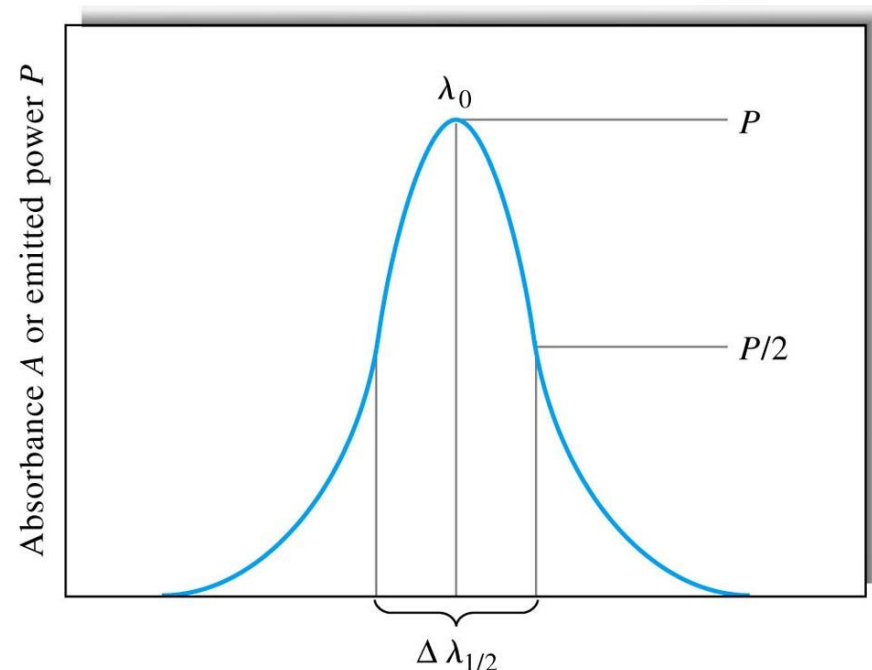
Closely spaced emission/absorption **lines** require high resolution monochromators to resolve fully.



# Atomic Line Widths

Theoretically, the discrete energies of atomic transitions would suggest a *line width* of *zero* (infinitely narrow).

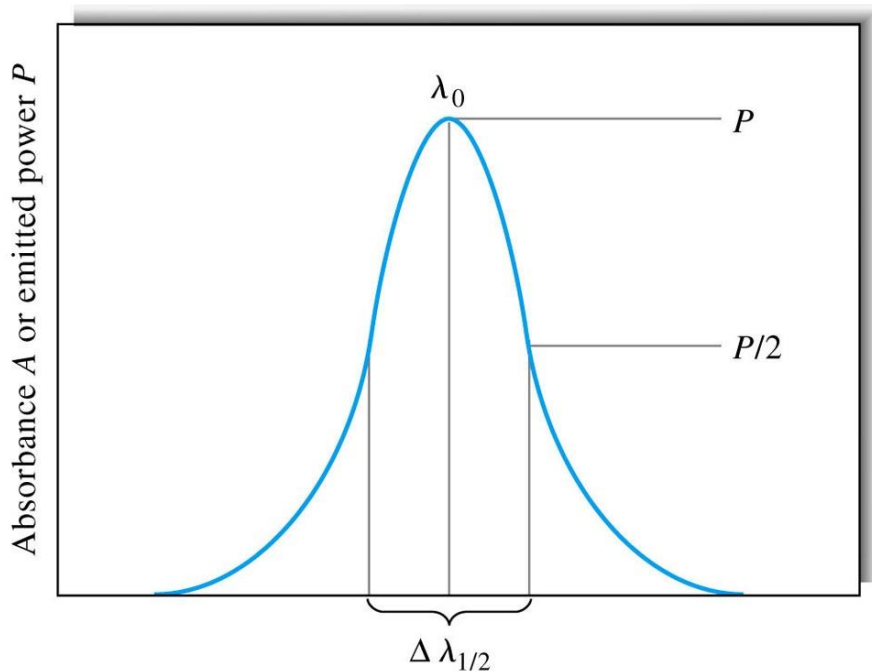
Several phenomena cause **line broadening** such that all atomic lines have finite widths (effective line width,  $\Delta\lambda_{1/2}$ ).



# Line Broadening – Uncertainty Effect

Finite lifetimes of atomic states leads to finite atomic line widths.

Because the lifetime of the upper and lower states of an atomic transition are finite there is an *uncertainty* in the energies of the states.



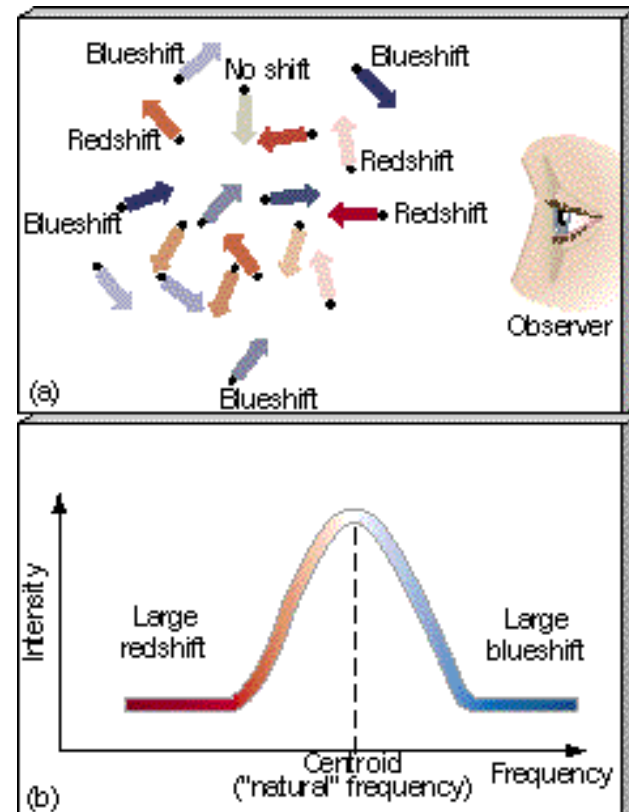
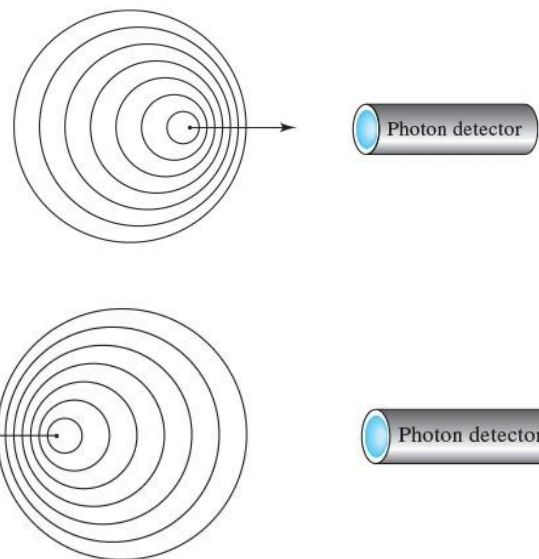
The lifetime of a ground-state electron is long, but short ( $10^{-7}$  to  $10^{-8}$  s) for the excited states.

Line widths due to uncertainty broadening are known as **natural line widths**.

# Line Broadening – Doppler Effect

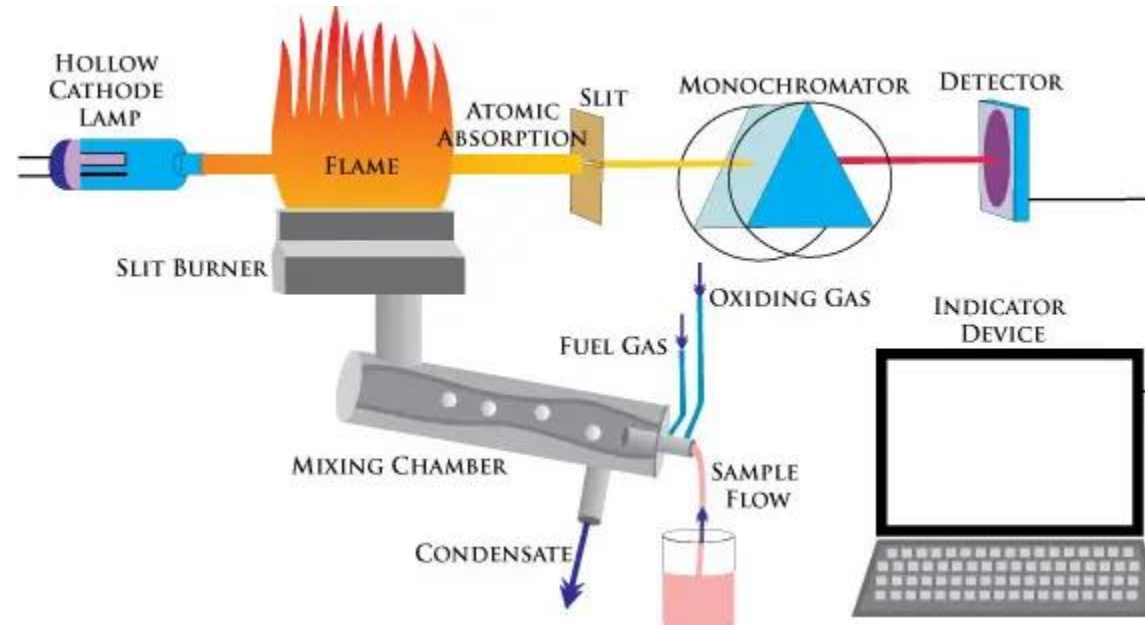
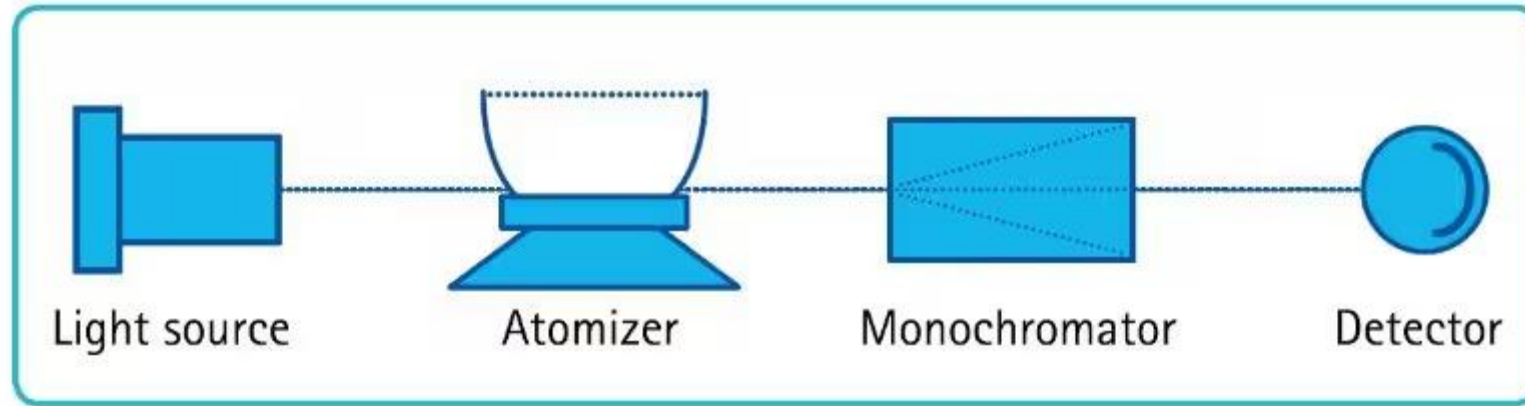
At very high temperature, the atoms move at very high velocities. The light emitted by the atoms will therefore be **Doppler-shifted**.

Since some atoms will be moving towards an observer and others away from them, the result will be that each line in the spectrum will be broadened.



- Maximum shift from atoms that move either towards or away from the transducer.
- No shift for atoms moving perpendicular to the transducer's path.
- Intermediate shifts for other atoms, depending on their speed and direction.

# Atomization



# Effects of Temperature on Atomic Spectra

**Temperature** has a significant effect on the ratio of atoms in the ground and excited states. as the temperature of atomization influences the number of atoms available in the excited state.

The magnitude of this effect is given by the *Boltzmann equation*:

$$\frac{N^*}{N_0} = \frac{g^*}{g_0} e^{\left(\frac{-\Delta E}{kT}\right)}$$

|            |                                                         |
|------------|---------------------------------------------------------|
| $N^*$      | number of atoms in the excited state                    |
| $N_0$      | number of atoms in the ground state                     |
| $g^*/g_0$  | statistical weights for the excited and ground states   |
| $\Delta E$ | energy difference between the excited and ground states |
| $k$        | Boltzmann constant ( $1.38 \times 10^{-23}$ J/K)        |
| $T$        | absolute temperature                                    |

# Practice Problem

Calculate the ratio of sodium atoms in the 3p excited states to the number in the ground state at 2500 and 2600 K (The statistical weights for the 3s and 3p quantum states are 2 and 6, respectively).

$$E = h\nu = h\frac{c}{\lambda} \quad E = 6.6261 \times 10^{-34} \text{ Js} \times \frac{2.988 \times 10^8 \text{ s}^{-1}}{589.3 \text{ nm} \times 10^{-9} \text{ m/nm}} = 3.37 \times 10^{-19} \text{ J}$$

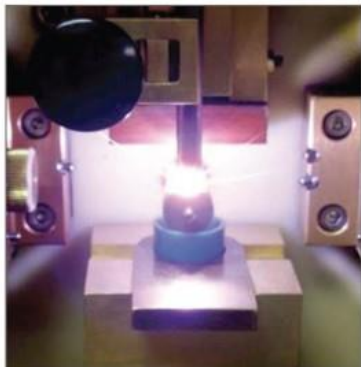
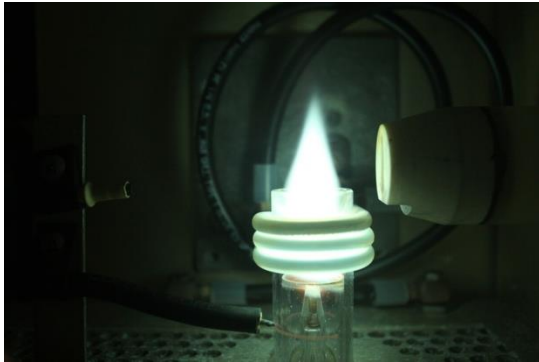
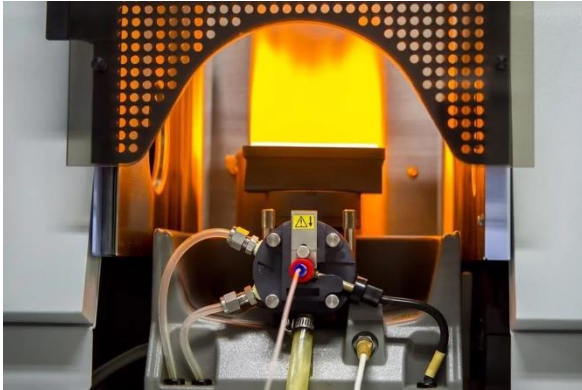
$$g^*/g_0 = 3$$

$$\frac{N^*}{N_0} = \frac{g^*}{g_0} e^{\left(\frac{-\Delta E}{kT}\right)} \quad \frac{N^*}{N_0} = 3e^{\left(\frac{-3.37 \times 10^{-19} \text{ J}}{(1.38 \times 10^{-23} \text{ J K}^{-1}) \times 2500 \text{ K}}\right)} = 1.72 \times 10^{-4}$$

$$\frac{N^*}{N_0} = 3e^{\left(\frac{-3.37 \times 10^{-19} \text{ J}}{(1.38 \times 10^{-23} \text{ J K}^{-1}) \times 2600 \text{ K}}\right)} = 2.50 \times 10^{-4}$$

Temperature fluctuation of only 100 K results in a 45% increase in the number of excited sodium atoms.

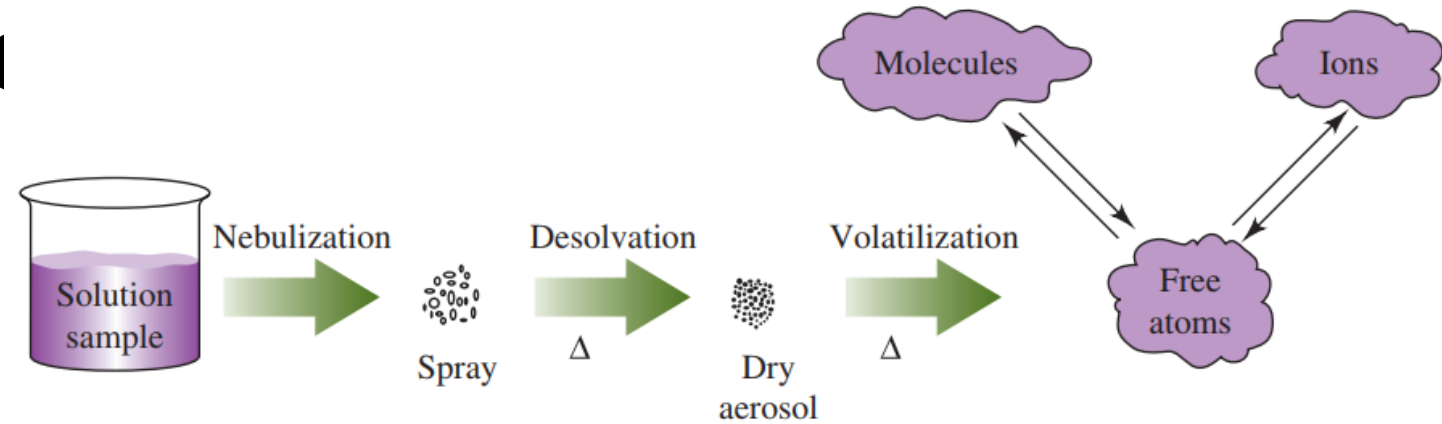
# Atomization Methods



| Atomizer                          | Typical Temperature (°C) | Temperature (K) | Sodium Excitation Likelihood |
|-----------------------------------|--------------------------|-----------------|------------------------------|
| Flame                             | 1700–3150                | 1973–3423 K     | Moderate                     |
| Electrothermal vaporization (ETV) | 1200–3000                | 1473–3273 K     | Low to moderate              |
| Inductively coupled plasma (ICP)  | 4000–6000                | 4273–6273 K     | High                         |
| Direct current plasma (DCP)       | 4000–6000                | 4273–6273 K     | High                         |
| Microwave-induced plasma (MIP)    | 2000–3000                | 2273–3273 K     | Moderate                     |
| Laser-induced plasma              | 8000–15,000              | 8273–15,273 K   | Very high                    |
| Electric arc                      | 4000–5000                | 4273–5273 K     | High                         |
| Electric spark                    | ~40,000 (?)              | ~40,273 K (?)   | Extremely high               |

# Sample Introduction

For all atomic spectroscopy measurements, the constituents of a sample must be converted into gaseous atoms and ions.



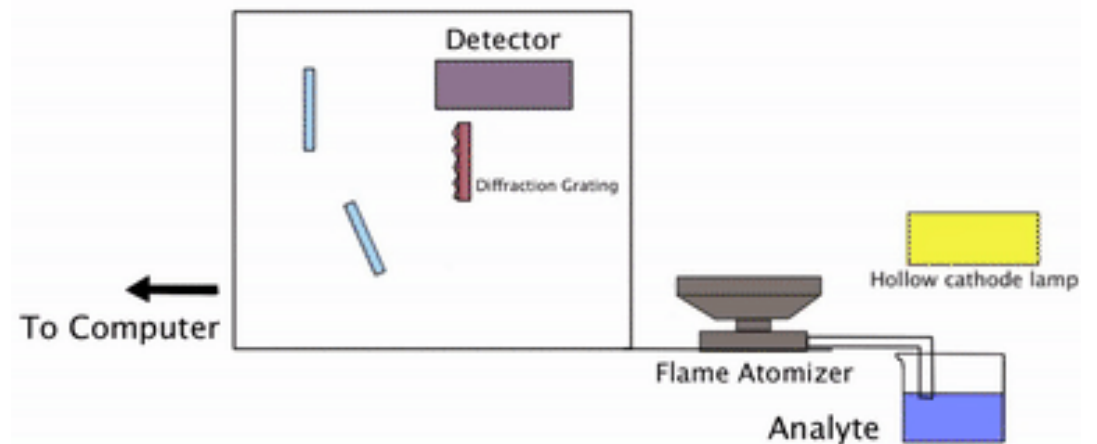
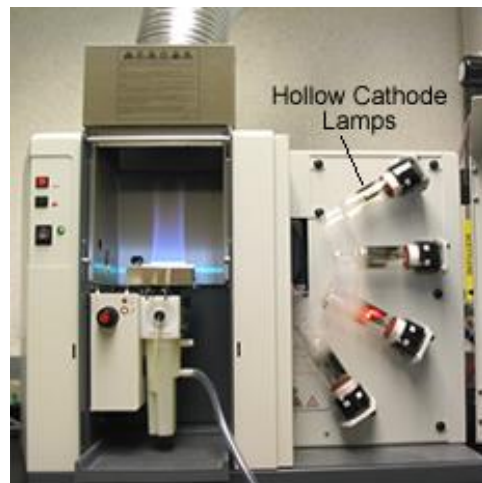
A reproducible and representative portion of the sample must first be introduced into the **atomizer**. Sample introduction in atomic spectroscopy measurements can limit the accuracy, precision, or detection limits of the method.

Introduction of the sample can be **continuous** or **discrete**.

Nebulizers are the most common method of liquid sample introduction for continuous atomizers.

| Method                      | Type of Sample             |
|-----------------------------|----------------------------|
| Pneumatic nebulization      | Solution or slurry         |
| Ultrasonic nebulization     | Solution                   |
| Electrothermal vaporization | Solid, liquid, or solution |
| Direct insertion            | Solid, powder              |
| Laser ablation              | Solid, metal               |
| Spark or arc ablation       | Conducting solid           |
| Glow-discharge sputtering   | Conducting solid           |

# Atomic Absorption Spectroscopy (AAS)

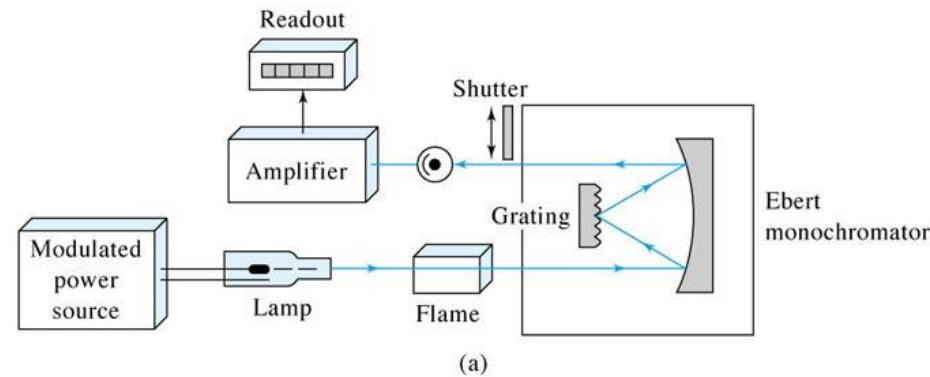


# AAS Spectrometers

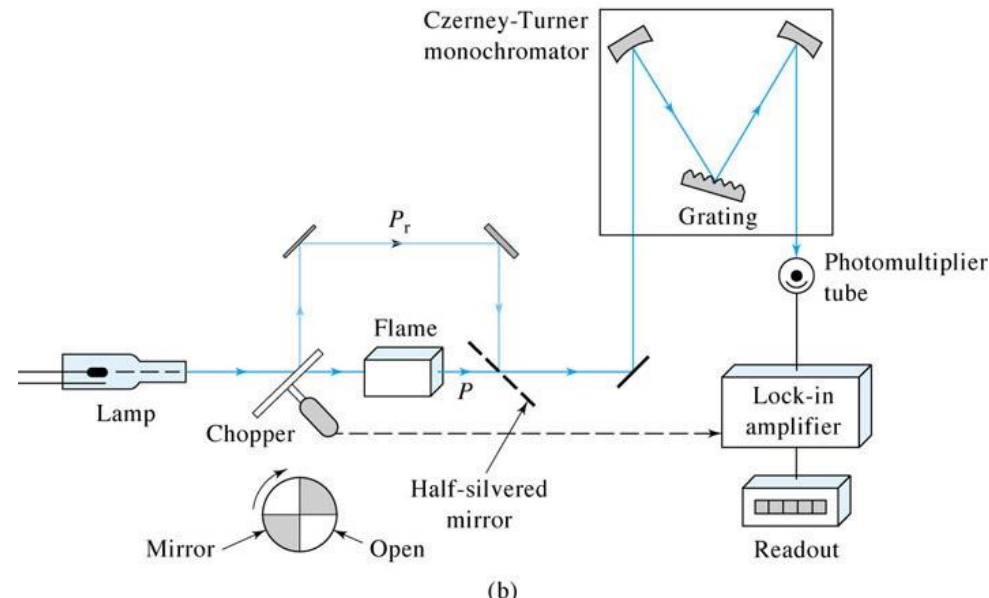
For atomic measurements, a sufficiently narrow bandwidth is required to isolate the analyte line from other lines that may interfere or reduce the measurement sensitivity.

Atomic absorption spectrometers consist of four main instrumental components:

- *Atomizer*
- *Radiation Source*
- *Wavelength selector*
- *Detector*



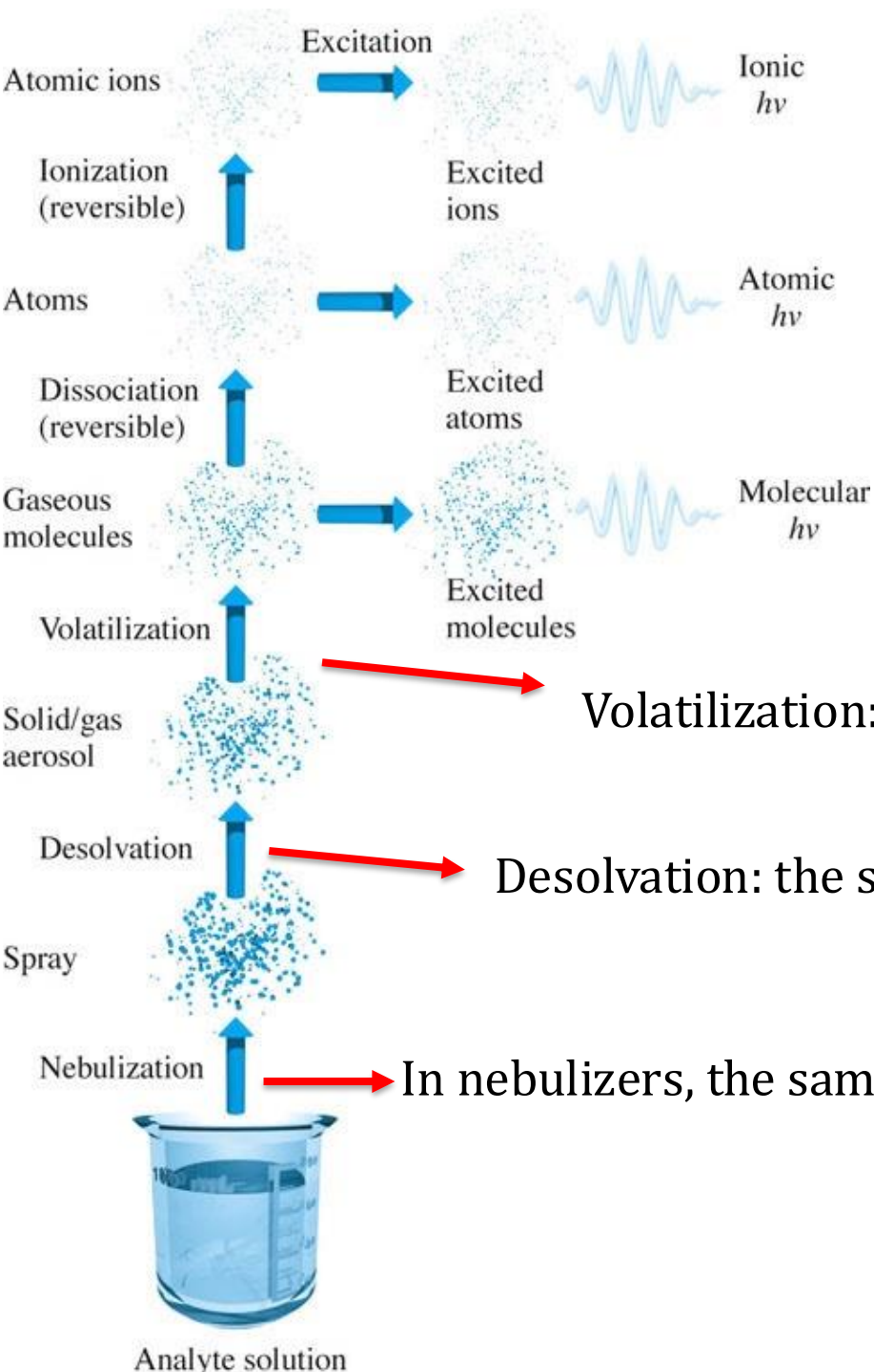
(a) Single-beam instruments



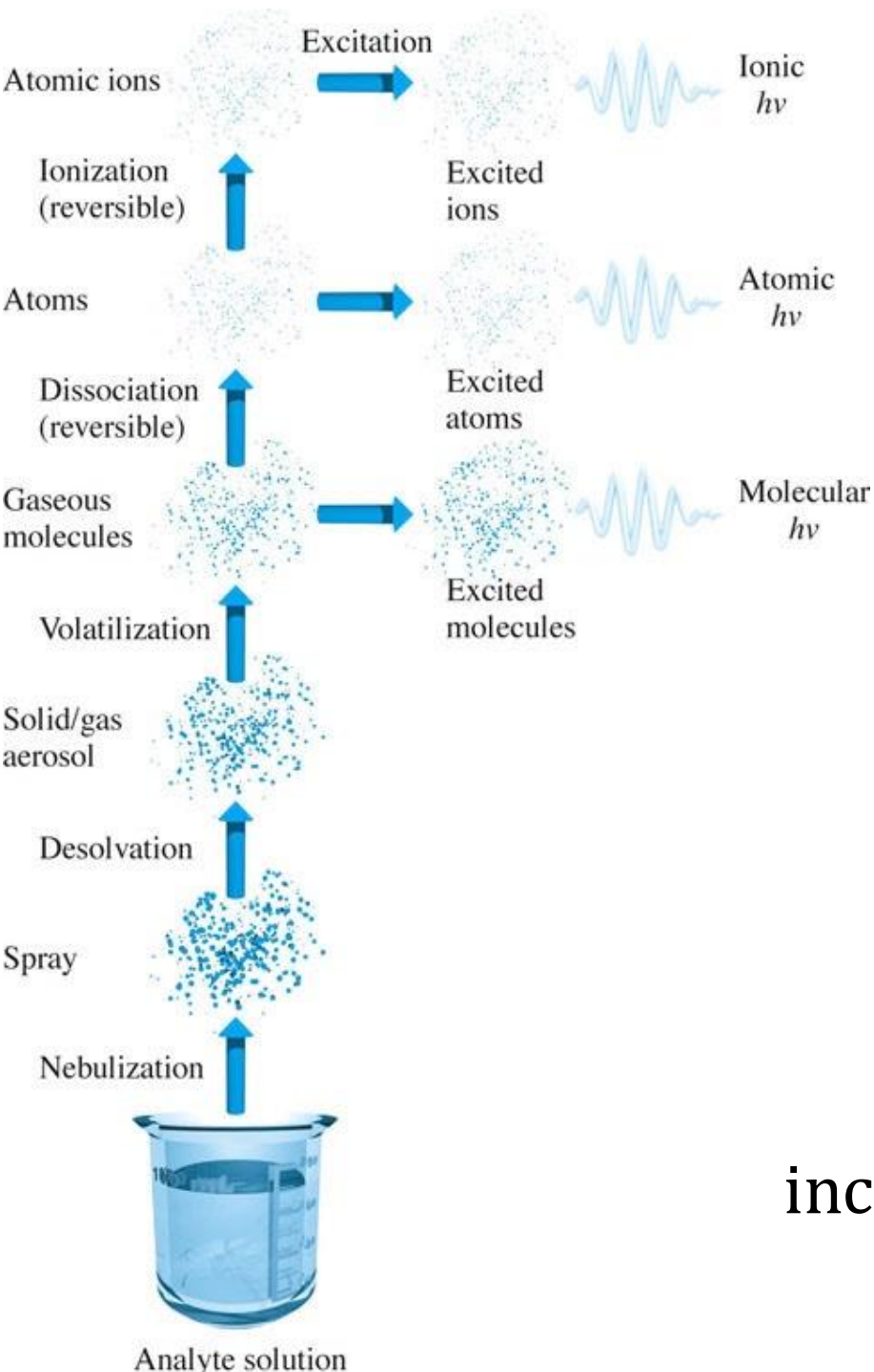
(b) Double-beam instruments

# Atomization

Atomization represents the most critical step in AAS – the sensitivity will be determined by the number of free atoms produced.



# Atomization



Dissociation: most of the molecules produces an atomic gas.  
Some of the atoms in the gas ionize to form cations and electrons.

Other molecules and atoms are produced in the flame as a result of interactions of the fuel with the oxidant and with the various species in the sample.

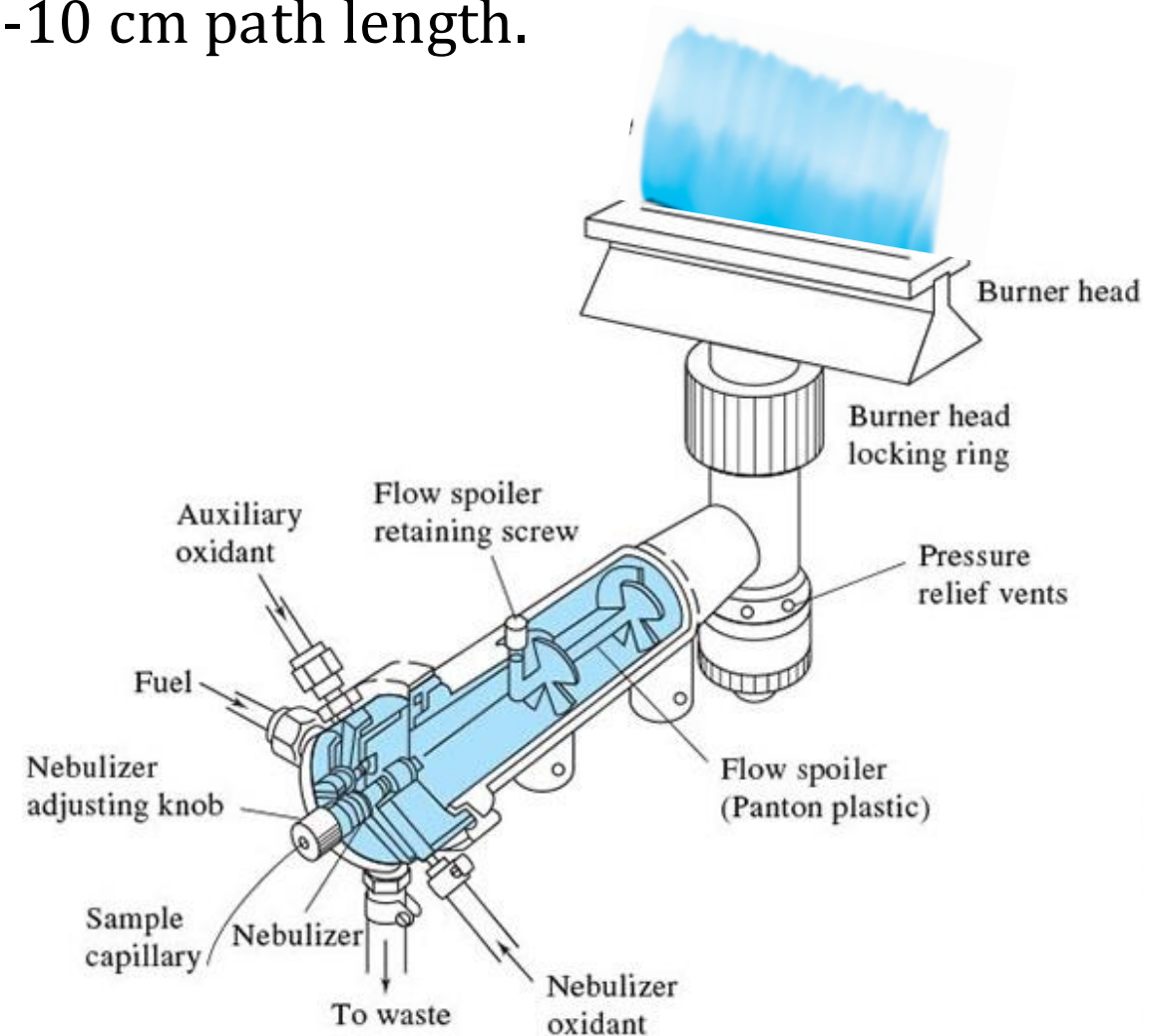
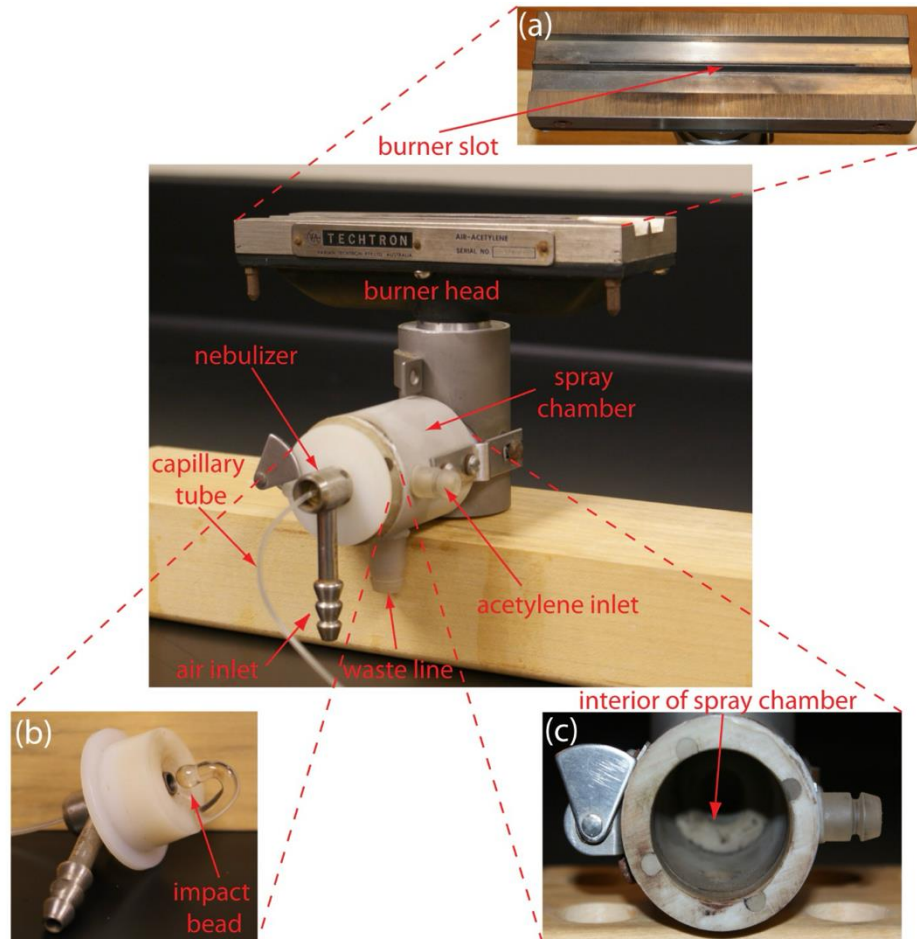
A fraction of the molecules, atoms, and ions are also excited by the heat of the flame to yield atomic, ionic, and molecular emission spectra



The most common methods of atomization include *flames (FAAS)*, *graphite furnaces (GFAAS)*, and *hydride generation (HGAAS)*.

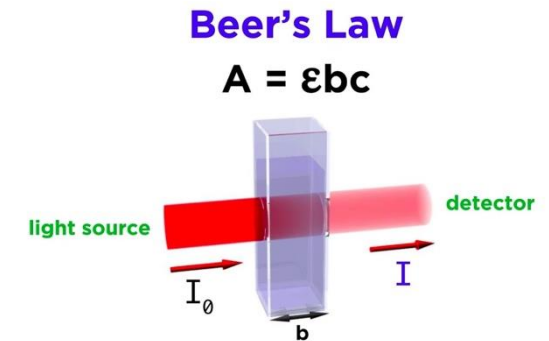
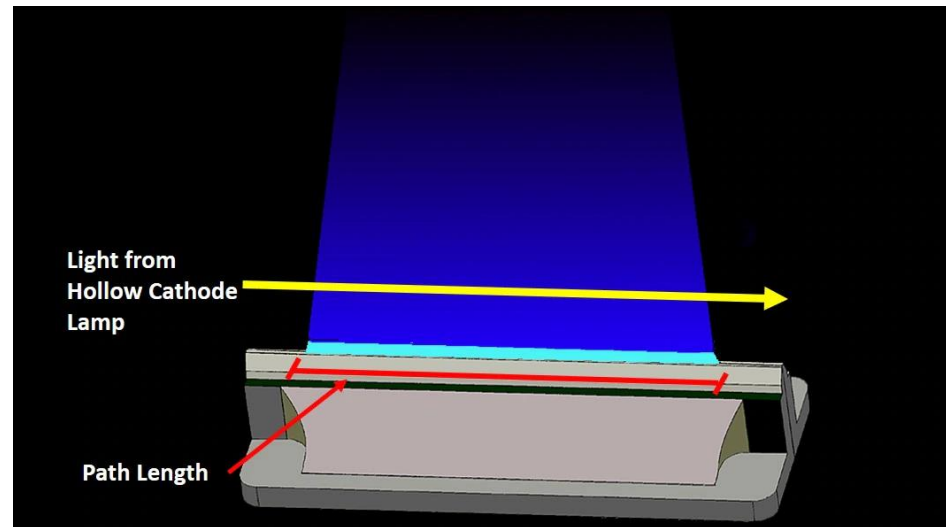
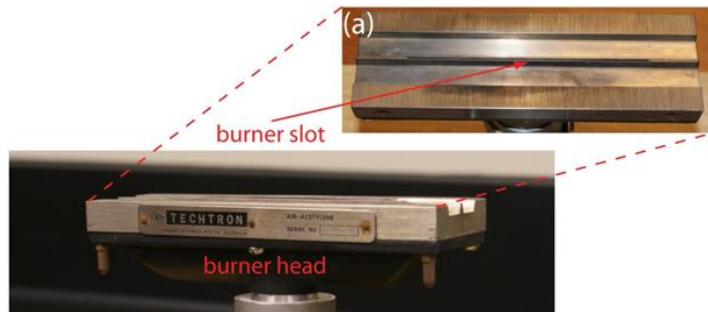
# Flame Atomization

The aerosol, formed by an oxidant flow, the oxidant, and fuel are mixed and burned in a slotted burner to produce a 5-10 cm path length.



# Flame Atomization

The slot burner provides a long optical pathlength and a stable flame. Absorbance increases linearly with the path length  $\rightarrow$  a long path length provides greater sensitivity.



# Flame Atomization

The *fuel/oxidant mixture* of the flame is chosen depending on the nature of the analyte and temperature required for atomization.

Elements which are difficult to decompose will required higher temperature flames.

| Fuel        | Oxidant       | Temperature (°C) | Maximum Burning Velocity (cm/s) |
|-------------|---------------|------------------|---------------------------------|
| Natural gas | Air           | 1700–1900        | 39–43                           |
| Natural gas | Oxygen        | 2700–2800        | 370–390                         |
| Hydrogen    | Air           | 2000–2100        | 300–440                         |
| Hydrogen    | Oxygen        | 2550–2700        | 900–1400                        |
| Acetylene   | Air           | 2100–2400        | 158–266                         |
| Acetylene   | Oxygen        | 3050–3150        | 1100–2480                       |
| Acetylene   | Nitrous oxide | 2600–2800        | 285                             |

- Each flame mixture will contribute to a background signal which must be subtracted during analysis.
- The gas mixture flow rate must exceed the burning velocity of the flame to avoid *flashback* (the flame propagation back into the burner).

# Flame Atomization

The *fuel/oxidant mixture* of the flame is chosen depending on the nature of the analyte and temperature required for atomization.

Fuel and oxidant are usually combined in approximately stoichiometric amounts. For the determination of metals that form *stable oxides*, however, a flame that contains an excess of fuel is often desirable.

| Fuel        | Oxidant       | Temperature (°C) | Maximum Burning Velocity (cm/s) |
|-------------|---------------|------------------|---------------------------------|
| Natural gas | Air           | 1700–1900        | 39–43                           |
| Natural gas | Oxygen        | 2700–2800        | 370–390                         |
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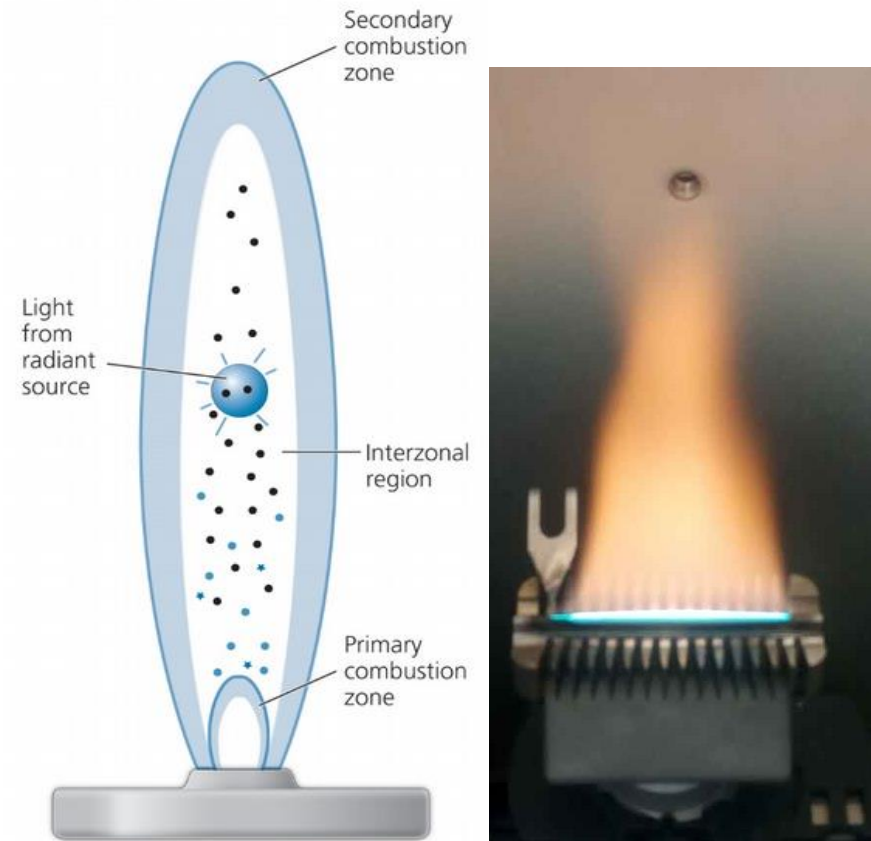
Flames can be of two types:

*Oxidizing flames:* fuel lean (oxidant rich)

*Reducing flames:* fuel rich

# Flame Atomization

Flame structure is dependent on the fuel/oxidant ratio, the types of fuel and oxidant used, and the type of burner.

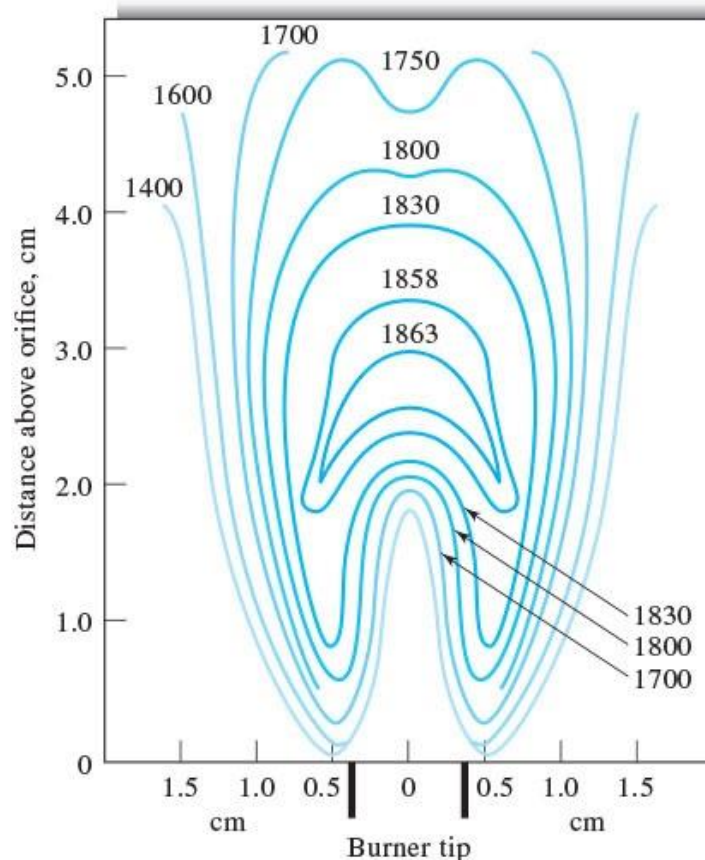
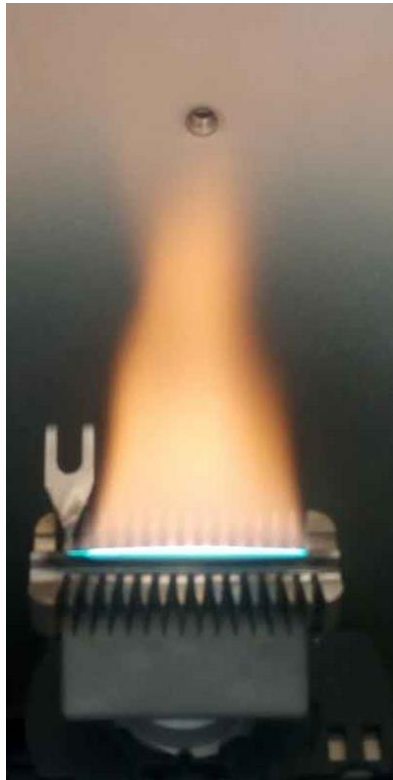
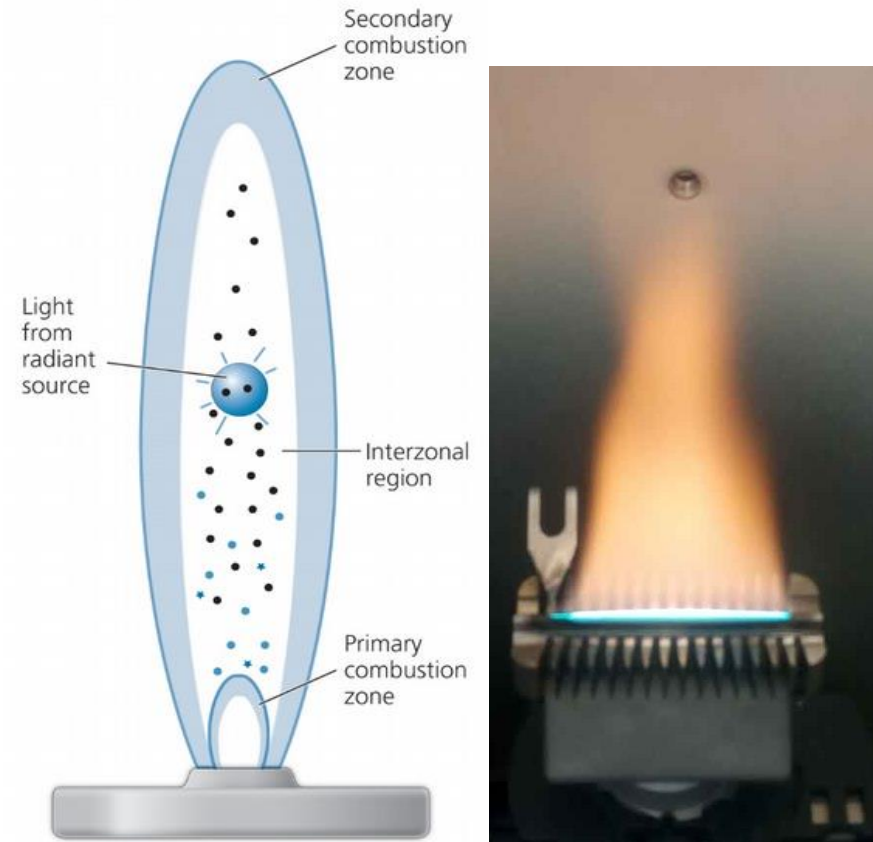


**The primary combustion:** The primary combustion zone of a hydrocarbon flame shows blue light from  $C_2$ ,  $CH$ , and other radicals. Thermal equilibrium is not reached here, so it's rarely used in spectroscopy.

**The interzonal area:** The interzonal region, narrow in stoichiometric flames but several cm tall in fuel-rich acetylene-oxygen or acetylene- $N_2O$  flames, contains many free atoms and is **the preferred zone for spectroscopy**.

# Flame Atomization

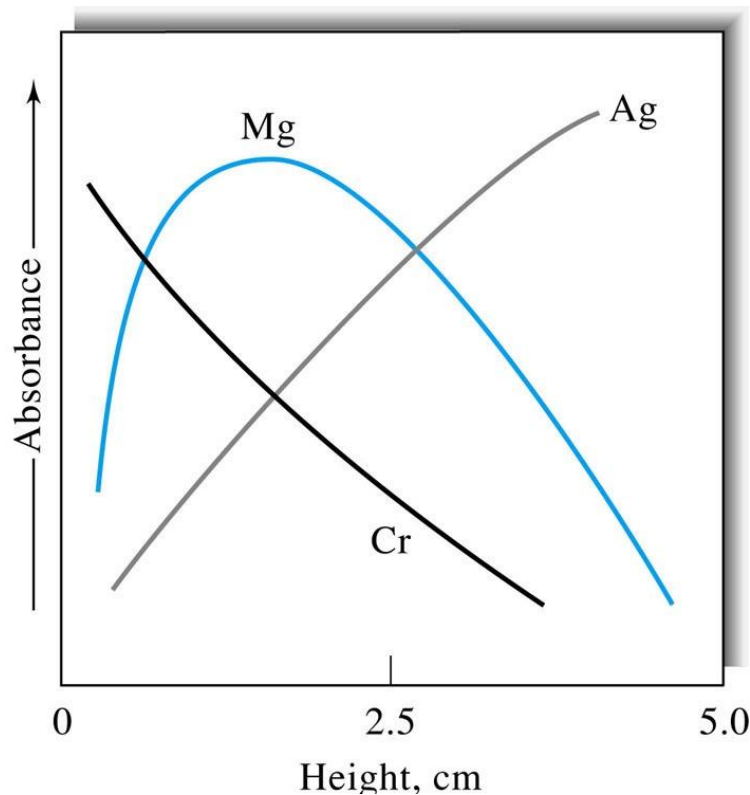
Temperature profiles for flames can be used in deciding where measurements should take place.



The maximum temperature is located in the flame about 2.5 cm above the primary combustion zone

# Flame Atomization

Elemental **absorption profiles** can similarly aid in selecting a flame height at which to measure.



*Mg:* Initial increase as the number of Mg atoms produced increases with longer exposure to the flame; decrease due to **formation of non-absorbing** oxides.

*Cr:* Highly prone to the **formation of stable oxides**, even at regions close to the burner.

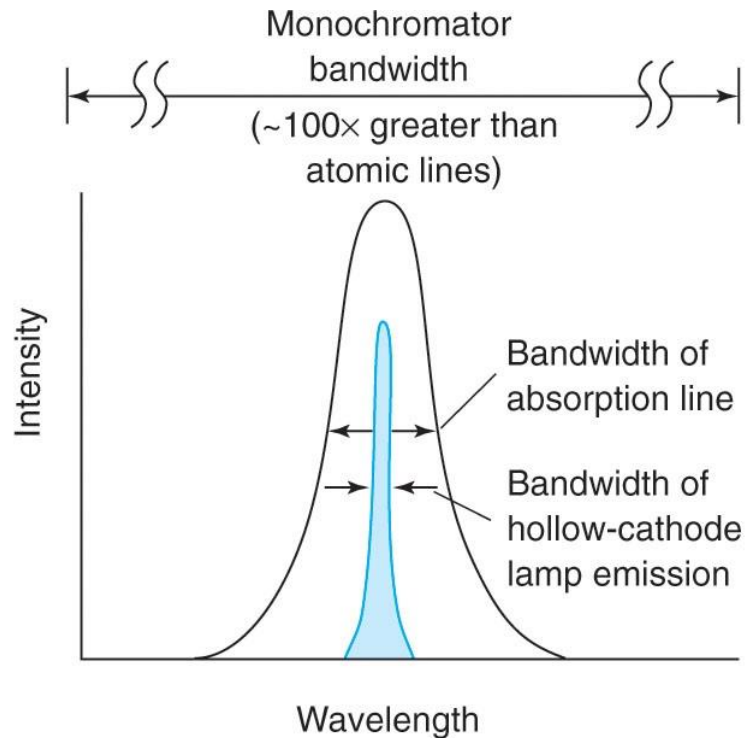
*Ag:* **Difficult to oxidize**, Ag shows a continuous increase in absorption due to additional atoms within the flame.

# Flame Atomization

| Advantages                                                                                                                                                                                           | Disadvantages                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>- Most reproducible of all liquid sample introduction methods.</li><li>- High elemental selectivity</li><li>- Fast analysis</li><li>- Easy operation</li></ul> | <ul style="list-style-type: none"><li>- Low sampling efficiency (95% of aspirated sample goes to waste).</li><li>- Lower sensitivity results</li><li>- Path length must be as long as possible to compensate</li><li>- Limited to the ppm detection range</li></ul> |

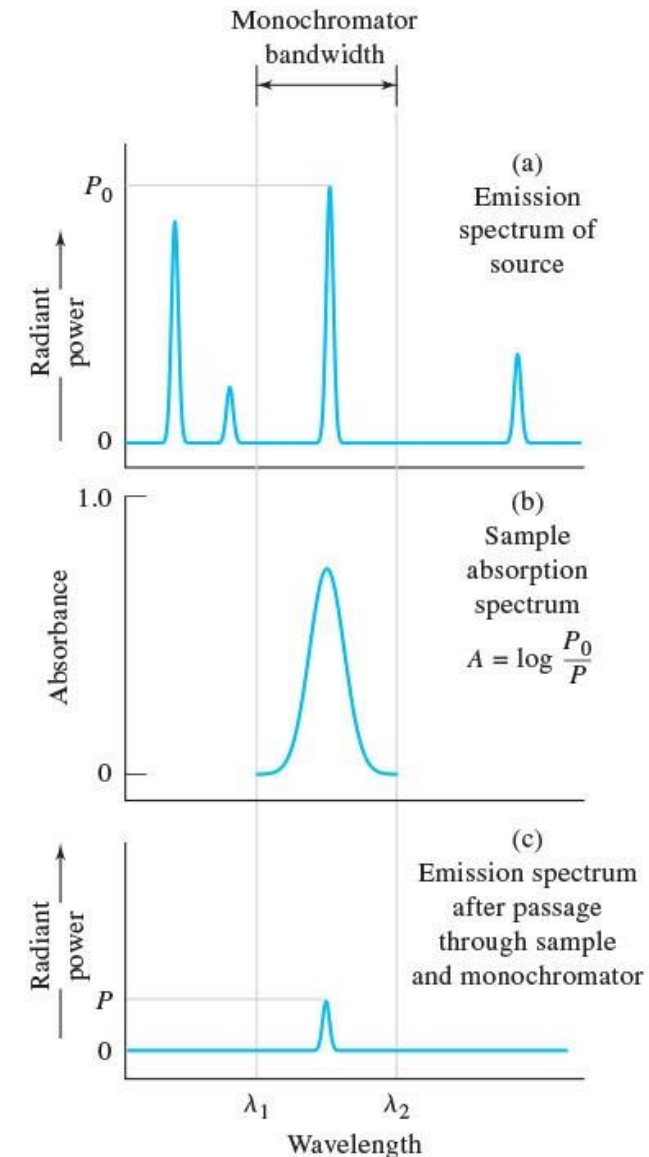
# Radiation sources

Atomic absorption methods can be highly specific due to the **unique and highly narrow absorption lines** (0.002 to 0.005 nm) of each element.



Line sources with bandwidths narrower than the absorption line width are often used.

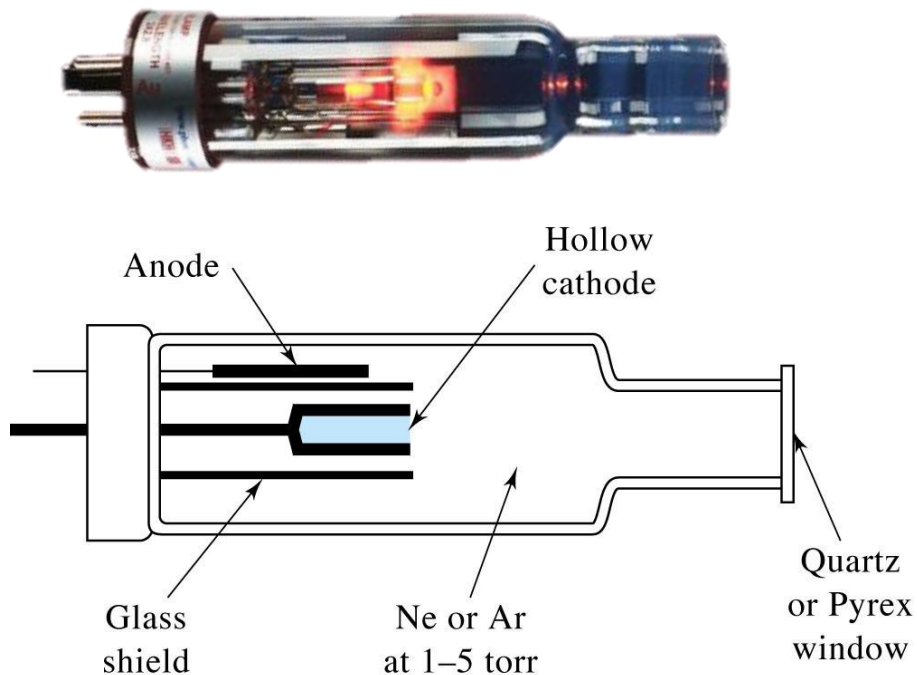
The disadvantage is that separate source lamps are required for each element of interest.



# Hollow Cathode Lamps

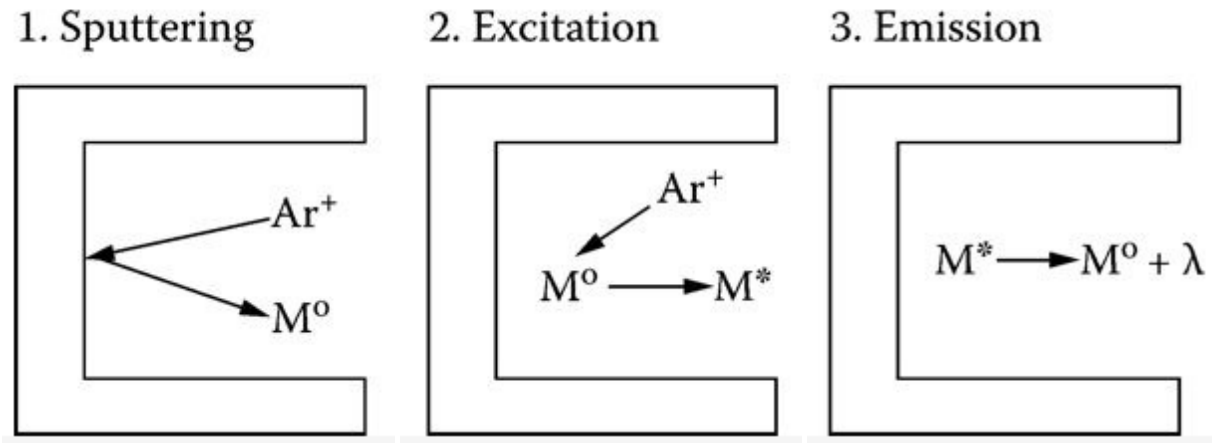
The **hollow cathode lamp** contains a tungsten anode and cylindrical cathode in a glass tube of low-pressure (1 to 5 torr) inert gas (Ar or Ne).

The cathode is formed from a hollowed-out cylinder or open cylinder of the metal whose spectrum will be emitted.

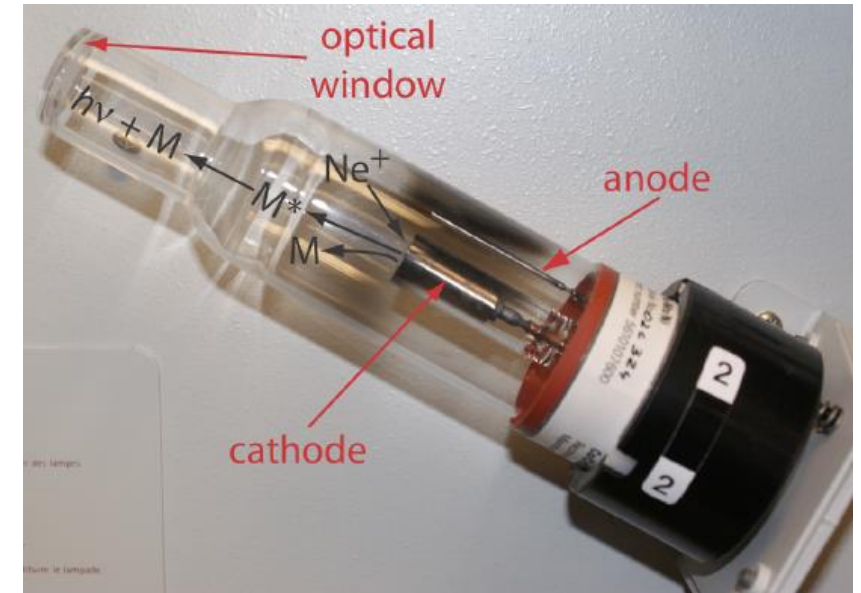


# Hollow Cathode Lamps

Application of a high voltage across the anode and cathode causes **atoms of the filler gas** to ionize and accelerate towards the cathode.



If the voltage is sufficiently large, the gaseous cations acquire enough kinetic energy to **dislodge some of the metal atoms** from the cathode surface and produce an atomic cloud in a process called sputtering.

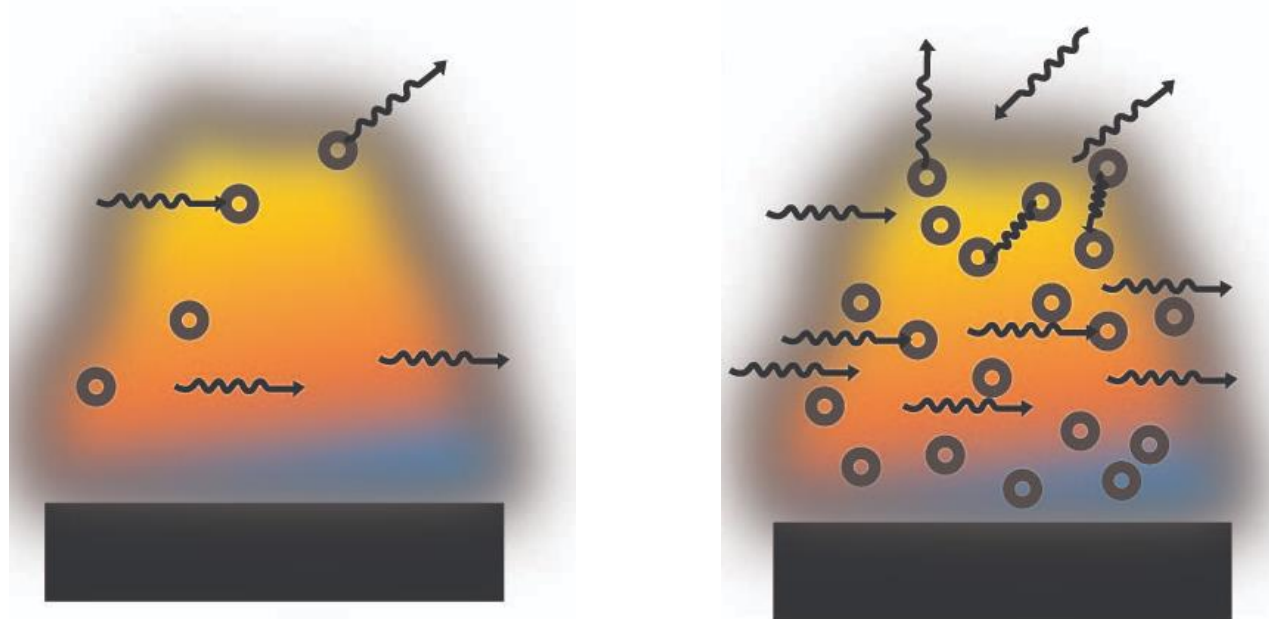


Eventually, the metal atoms diffuse back to the cathode surface or to the glass walls of the tube and are redeposited

# Hollow Cathode Lamps

The efficiency of the hollow cathode source depends on its geometry as well as the operating voltage.

**Self-absorption:** the greater currents produce an increased number of unexcited atoms in the cloud, which are capable of absorbing the radiation emitted by the excited ones, leads to lowered intensities.



# Applications of AAS

**TABLE 9-3** Detection Limits (ng/mL)<sup>a</sup>  
for Selected Elements

| Element | AAS<br>Flame | AAS<br>Electro-<br>thermal | AES<br>Flame | AES<br>ICP | AFS<br>ICP |
|---------|--------------|----------------------------|--------------|------------|------------|
| Al      | 30           | 0.1                        | 5            | 0.2        | 5          |
| As      | 200          | 0.5                        | —            | 2          | 15         |
| Ca      | 1            | 0.25                       | 0.1          | 0.0001     | 0.4        |
| Cd      | 1            | 0.01                       | 2000         | 0.07       | 0.1        |
| Cr      | 4            | 0.03                       | 5            | 0.08       | 0.6        |
| Cu      | 2            | 0.05                       | 10           | 0.04       | 0.2        |
| Fe      | 6            | 0.25                       | 50           | 0.09       | 0.3        |
| Hg      | 500          | 5                          | —            | —          | 5          |
| Mg      | 0.2          | 0.002                      | 5            | 0.003      | 0.3        |
| Mn      | 2            | 0.01                       | —            | 0.01       | 1          |
| Mo      | 5            | 0.5                        | 100          | 0.2        | 8          |
| Na      | 0.2          | 0.02                       | 0.1          | 0.1        | 0.3        |
| Ni      | 3            | 0.5                        | 600          | 0.2        | 0.4        |
| Pb      | 8            | 0.1                        | 200          | 1          | 5          |
| Sn      | 15           | 5                          | 300          | —          | 200        |
| V       | 25           | 1                          | 200          | 0.06       | 25         |
| Zn      | 1            | 0.005                      | 50000        | 0.1        | 0.1        |

The detection limits for AAS with flame atomization lie in the range of 1 to 20 ng/mL (0.001 to 0.020 ppm).



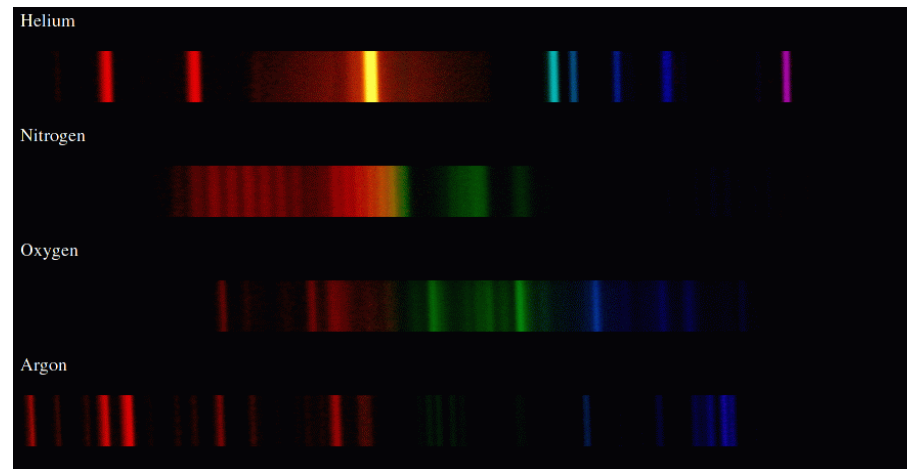
AAS: atomic absorption spectroscopy  
AES: atomic emission spectroscopy  
AFS: atomic fluorescence spectroscopy  
ICP: inductively coupled plasma

# Atomic Absorption Spectroscopy (AAS)

| Advantages                                                                                                                                                                                                                                                                               | Disadvantages                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>- Measures ~70 elements in almost any sample.</li><li>- Analysis is independent of the analytes chemical form</li><li>- Trace level analysis (ppm, ppb)</li><li>- Rapid analysis</li><li>- Little interference of elemental absorptions.</li></ul> | <ul style="list-style-type: none"><li>- Not applicable for structural analysis</li><li>- Not applicable for speciation analysis</li><li>- Sometimes, only one element can be analyzed at a time due to the specificity of the radiation source</li></ul> |

AAS is almost exclusively used for **quantitative analysis** of elements!

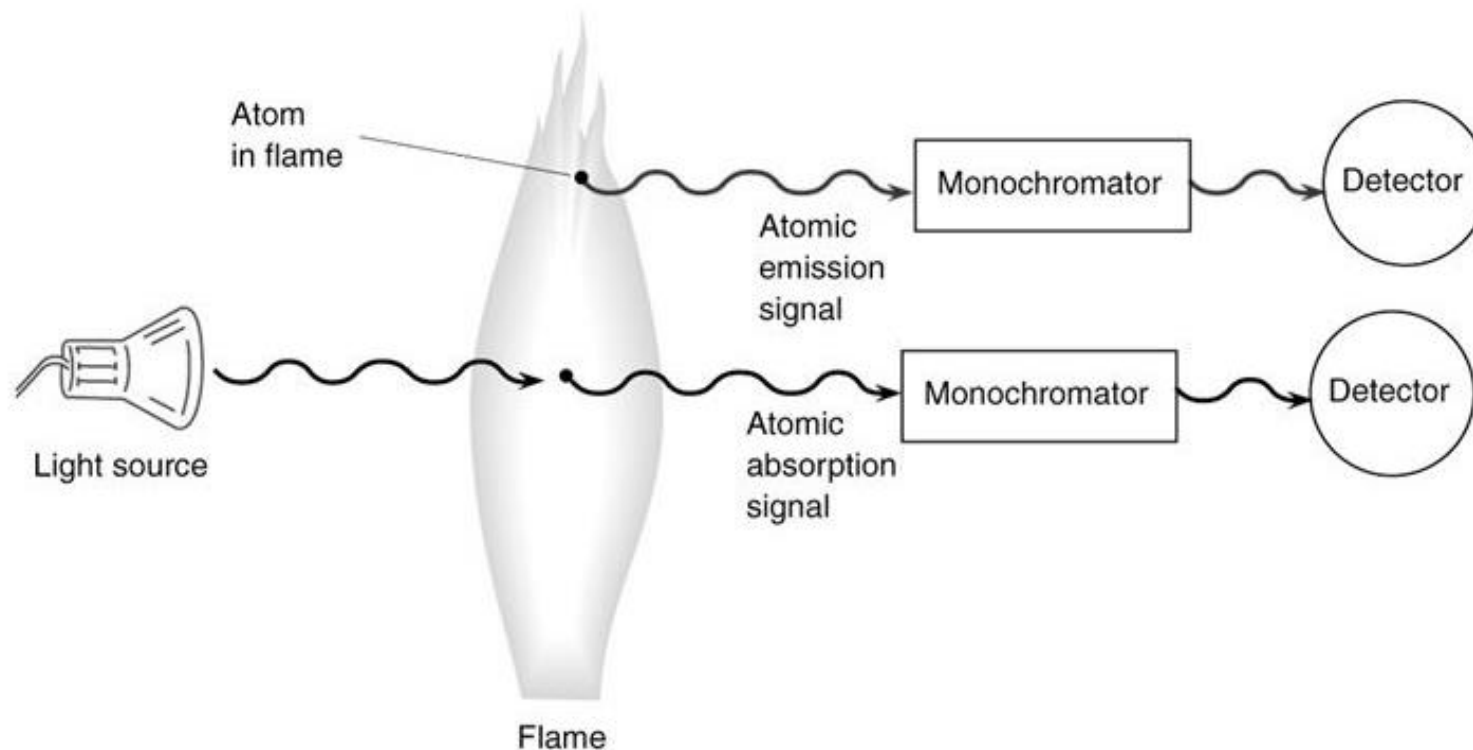
# Atomic Emission Spectroscopy (AES)



# Atomic Emission Spectroscopy (AES)

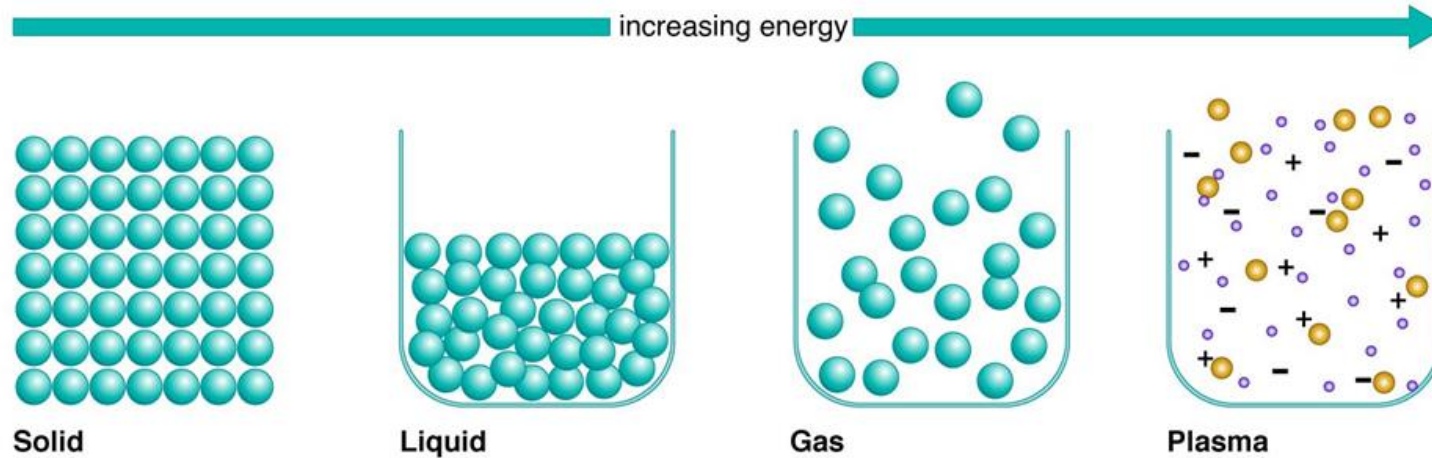
In atomic emission measurements, the *atomizer becomes the source* of excitation as well as atomization.

**Plasmas** are the most important and widely used sources in AES.



# Plasma: The Fourth State of Matter

Plasma is superheated matter: so hot that the electrons are ripped away from the atoms forming an ionized gas.



In another way: just as a liquid will boil, changing into a gas when energy is added, heating a gas will form a plasma: **a soup of positively charged particles (ions) and negatively charged particles (electrons).**



# Plasmas in Atomic Emission Spectroscopy

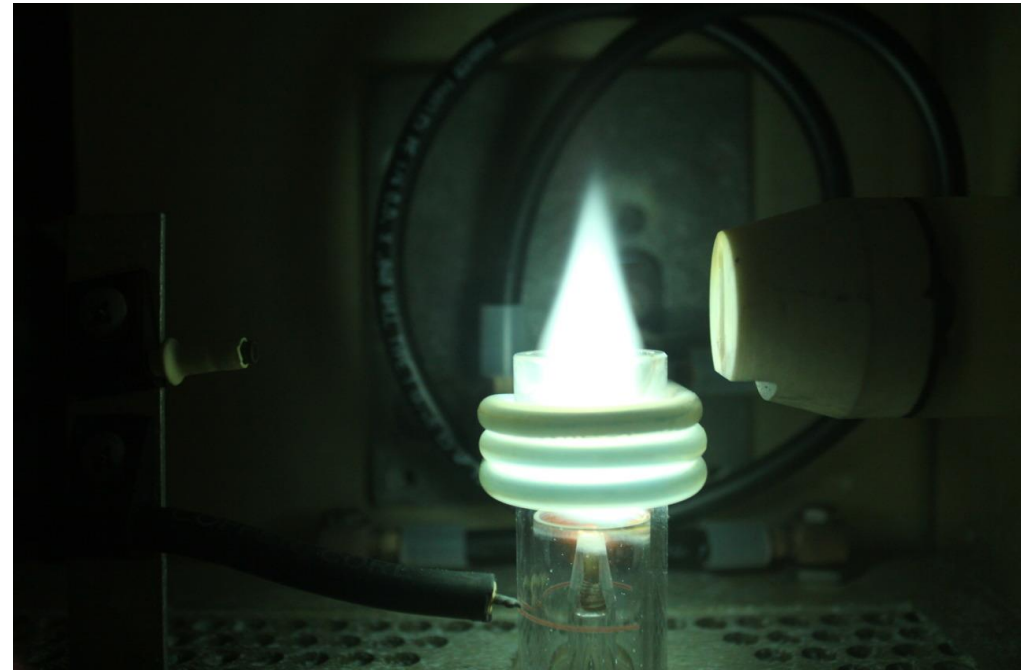
One of the most commonly used plasmas is composed of Ar.

Argon ions, once formed in a plasma, can absorb sufficient power from an external source to maintain the temperature at a level where further ionization sustains the plasma indefinitely.

Such plasmas achieve temperatures as high as **10,000 K**.

Two main plasma-based techniques:

- inductively coupled plasma (ICP)
- microwave-induced plasma (MIP)



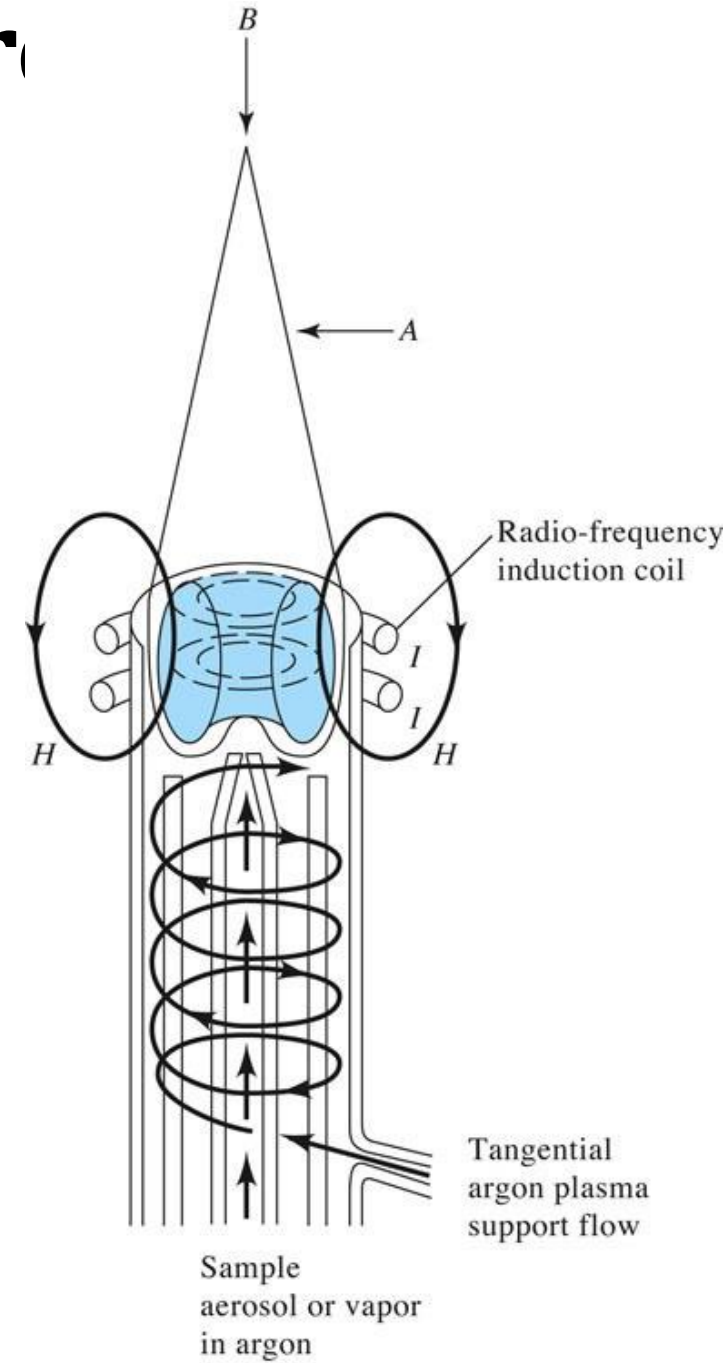
# Inductively Coupled Plasma Source



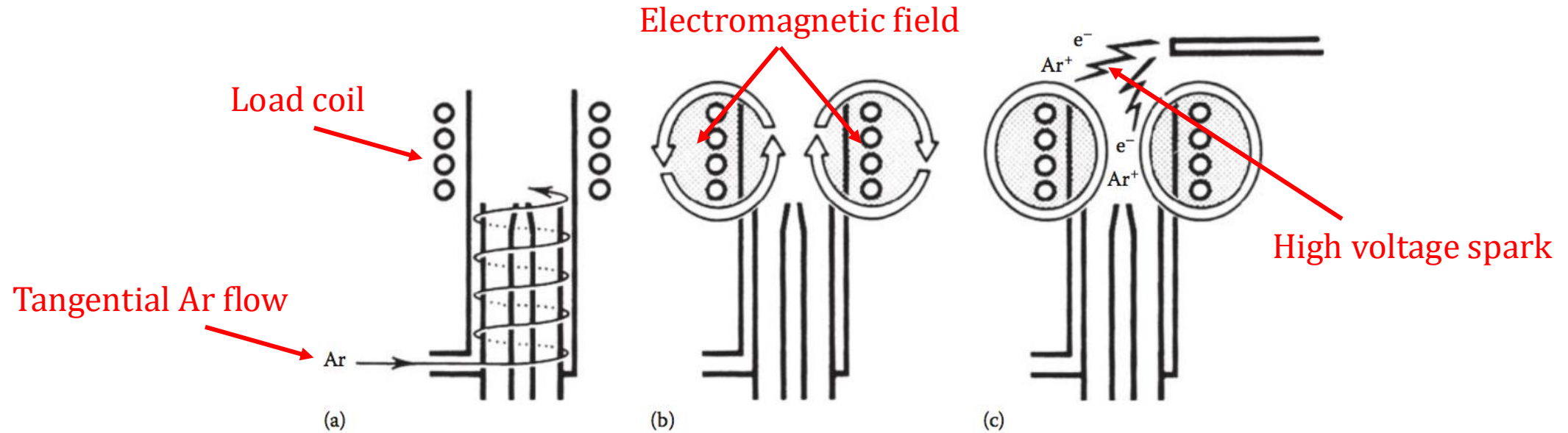
The **ICP torch** consists of three concentric quartz tubes through which the argon gas flows.

Surrounding the top of the torch is a water-cooled induction coil powered by a radio-frequency (RF) generator.

As the plasma is composed of highly energetic, ionized inert gases, it is **electronically conductive** and will interact with the **fluctuating magnetic field** (H).

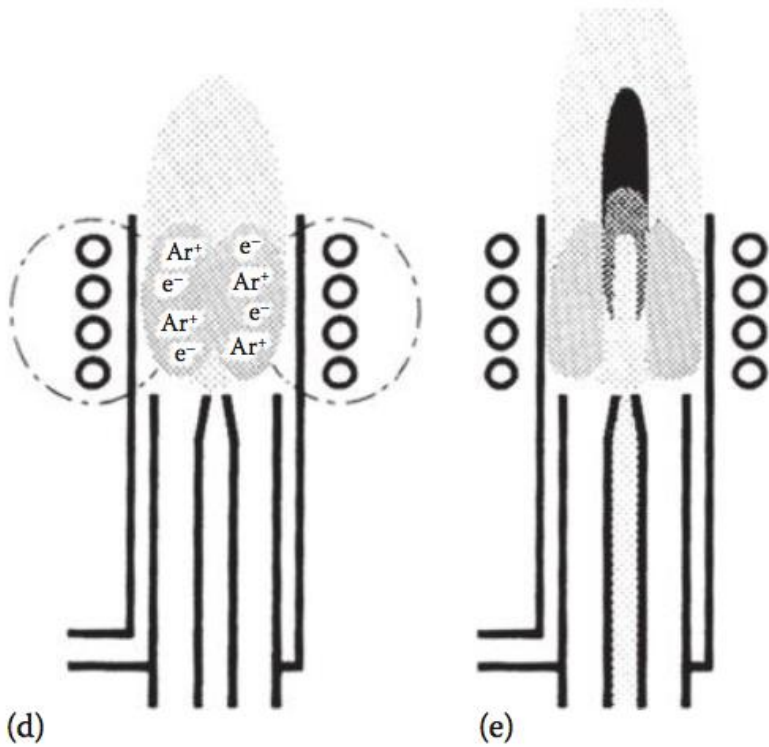


# Inductively Coupled Plasma Source



- Argon gas is swirled through the torch.
- RF power is applied to the induction (load) coil, inducing an electromagnetic field.
- Ionization of the argon gas is initiated via a spark from a Tesla coil.

# Inductively Coupled Plasma Source



- d) The ions and their associated electrons produced via the spark now interact with the fluctuating magnetic field.
- Acceleration of the electrons by the RF field causes **more ionization** and the formation of a plasma.
  - The plasma is heated via the **ohmic resistance** of ions and electrons within the coil flowing in a **closed annular path**.
- e) Nebulizer flow carries the sample aerosol into the plasma.

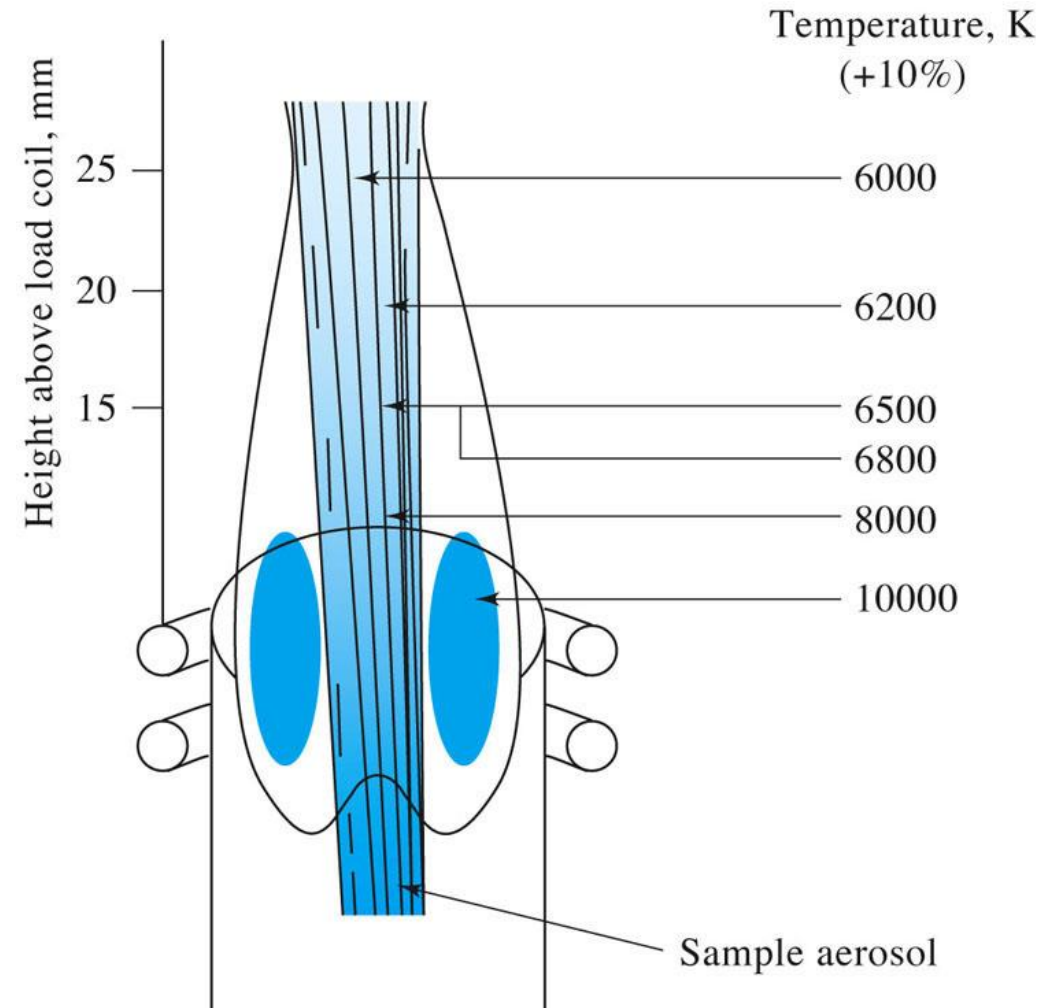
# Inductively Coupled Plasma Source

Temperatures of 5500 – 8000 K and an increased residence time result in a more complete atomization than AAS and reduced chemical interferences.

- *Plasma Core:* few mm above coil
- *Spectral Region:* 15-20 mm above coil



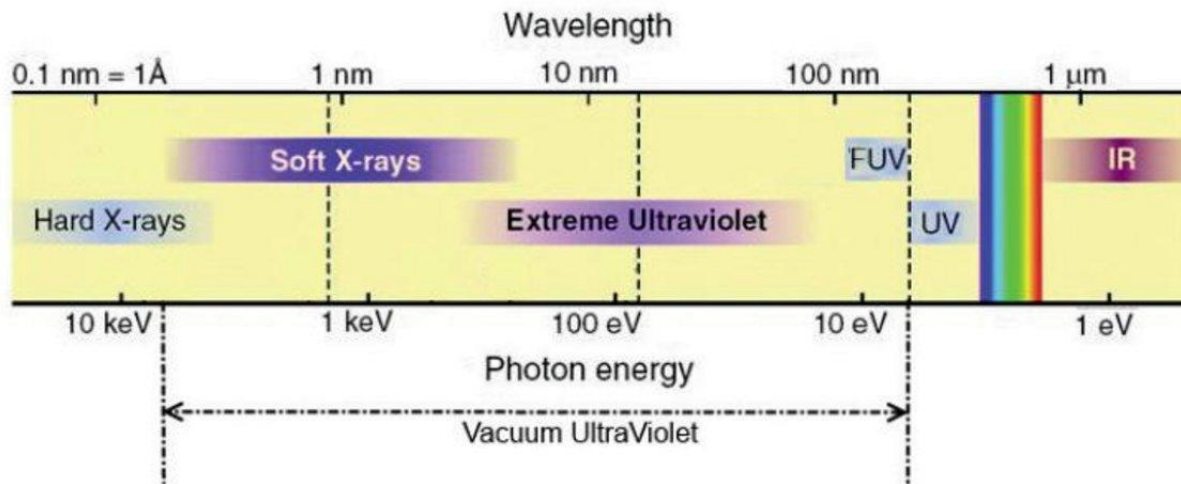
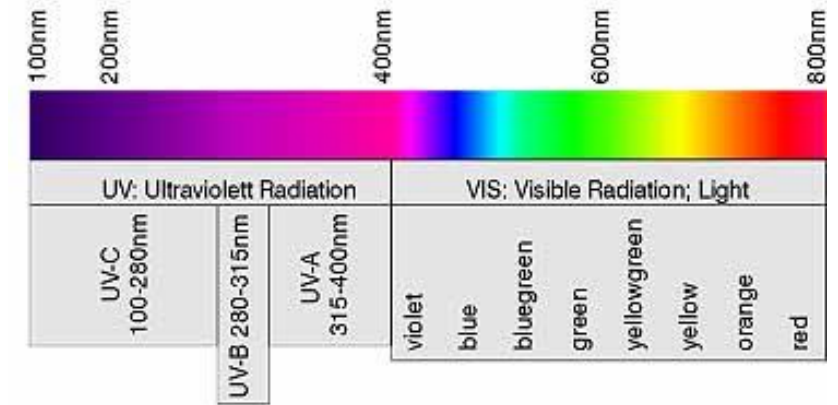
[link](#)



# Plasma AES Spectrometers

Most conventional AES instruments encompass the UV-Vis region from ~190 to 800 nm.

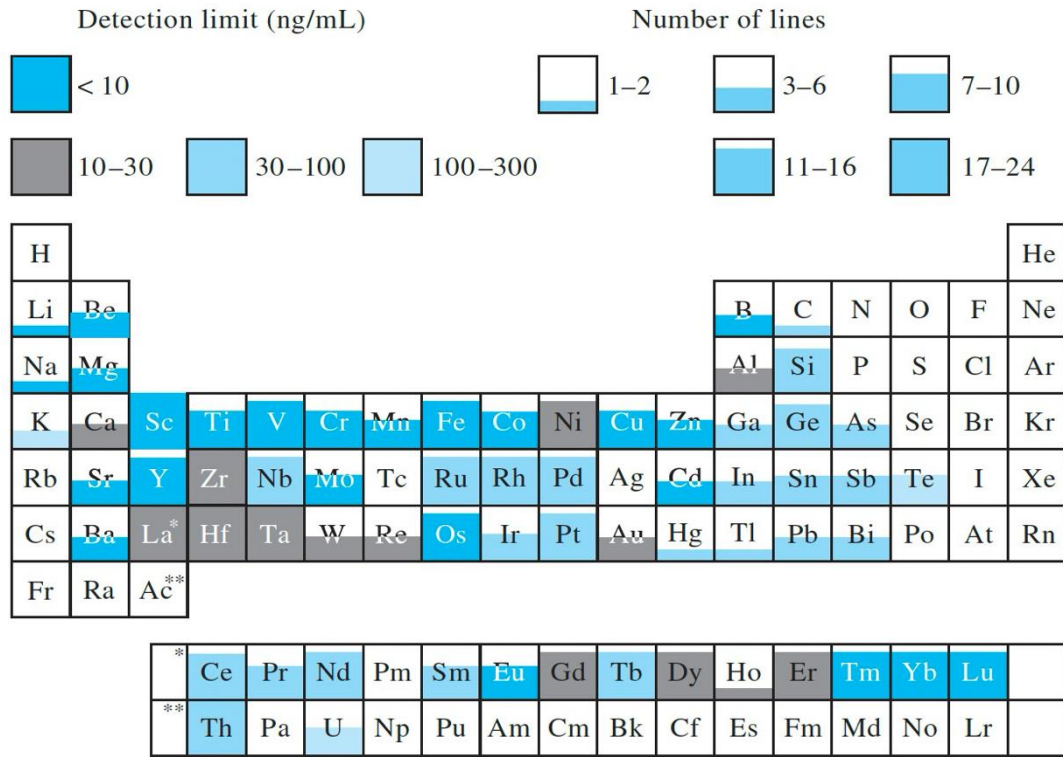
In air, oxygen and nitrogen strongly absorb anything below about 190 nm.



Some are capable of vacuum UV detection (vacuum emission spectroscopy (VES)) for above ~150 – 160 nm.

- The Far Ultraviolet (FUV): 200-120 nm
- The Extreme Ultraviolet (EUV): 120-10 nm
- The Soft X-Rays (Soft X): 15-0.2 nm

# Elemental Analysis with AES



In principle, all elements can be determined by plasma emission spectroscopy.

However, more practically it can determine 60 elements.

- Elements like B, P, N, S, and C need vacuum spectrometers because their lines are <180 nm, where air absorbs.
- Alkali metals (Li, K, Rb, Cs) emit in the near-IR, which UV-optimized plasma spectrometers detect poorly.

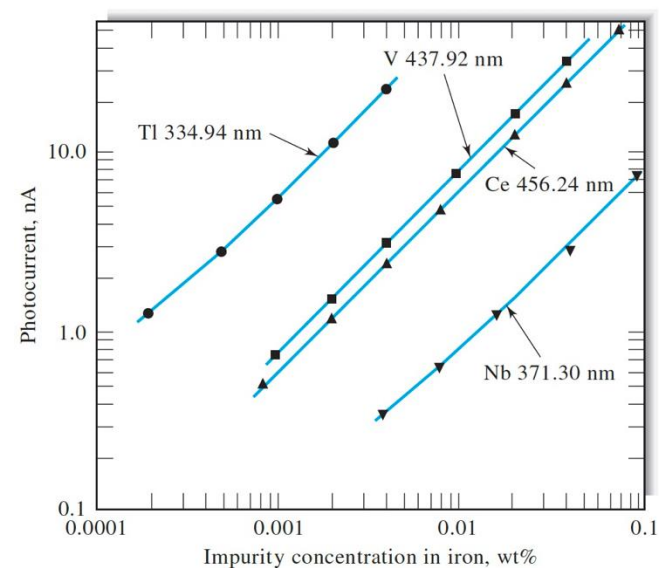
The more atomic lines present, the more likely a suitable line free of spectral interference will be found.

Line selection is based on what other elements may be present in the sample and whether it is likely line overlap will occur.

# Quantitative Analysis

## Interferences:

- Chemical interferences and matrix effects are significantly lower with plasma sources than with other atomizers.
- At low analyte concentrations, the background emission due to recombination of argon ions with electrons is large enough to require careful corrections.



*log-log plots*

A major cause of nonlinearity is self-absorption (only at high analyte concentrations), in which the output signal is reduced because of absorption by ground-state atoms in the medium.

**TABLE 10-3** Comparison of Detection Limits for Several Atomic Spectral Methods

| Method                    | Number of Elements Detected at Concentrations of |          |            |             |          |
|---------------------------|--------------------------------------------------|----------|------------|-------------|----------|
|                           | <1 ppb                                           | 1–10 ppb | 11–100 ppb | 101–500 ppb | >500 ppb |
| ICP emission              | 9                                                | 32       | 14         | 6           | 0        |
| Flame atomic emission     | 4                                                | 12       | 19         | 6           | 19       |
| Flame atomic fluorescence | 4                                                | 14       | 16         | 4           | 6        |
| Flame atomic absorption   | 1                                                | 14       | 25         | 3           | 14       |

Data abstracted with permission from V. A. Fassel and R. N. Kniseley, *Anal. Chem.*, 1974, 46 (13), 1110A. Copyright 1974 American Chemical Society.

Detection limits correspond to a signal that is twice as great as the standard deviation for the background noise.

# Atomic Emission Spectroscopy (AES)

| Advantages                                                                                                                                                                                                                                                                                                                                                                                                | Disadvantages                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>- Lower susceptibility to chemical interferences due to higher operating temperatures than AAS</li><li>- Easily quantitate refractory elements</li><li>- Single excitation conditions can be used for a wide range of elements</li><li>- Simultaneous emission spectrum recording</li><li>- Determination of non-metals</li><li>- Increased linear ranges</li></ul> | <ul style="list-style-type: none"><li>- Increased spectral interferences due to large quantity of emission lines produced by high temperature source</li><li>- Requires higher resolution instrumentation to differentiate atomic lines</li><li>- More expensive than AAS methods</li><li>- Requires higher operator skill</li></ul> |